

# Context Shift and Viewpoint Shift: Super Monsters Revisited\*

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**Abstract.** While Kaplan's classic theory of indexicals took them to be rigidly anchored to the context of the actual speech act, more recent research has found multiple cases in which an indexical depends on the context of a reported speech act, suggesting that attitude verbs can sometimes shift the context (in Kaplan's terms, attitude operators can be "monsters"). It was further proposed that context shift is sometimes overtly realized in sign language by Role Shift, an operation in which the signer visibly adopts someone else's perspective. However, in ASL and LSF, Role Shift involves peculiarities that have motivated the development of non-standard context-shifting operators ('Super Monsters'), and also of alternative treatments (in terms of demonstrations). First, Role Shift arises not just in attitude reports but also in action reports, where no monsters have been found in spoken language. Second, Role Shift forces elements within its scope to be interpreted in a maximally iconic fashion ('Maximize Iconicity'), something that has not been found in spoken language monsters. We argue that a key observation, familiar from informal descriptions, has been missed by formal analyses: Role Shift usually shifts the viewpoint ('camera position') relative to which iconic elements are evaluated. Building on the formal framework of Iconological Semantics, which has an explicit pictorial component, we develop a minimal analysis in which Role Shift only involves overt viewpoint shift. We find that it arguably leaves some facts unaccounted for, and consider an alternative in which Role Shift involves both context shift and viewpoint shift, hence a new take on Super Monsters. While our main new data are from ASL, we include gestural examples that suggest that the analysis could be extended to speech with gestures.

Keywords: context shift, viewpoint shift, sign language semantics, ASL, iconicity, pictorial semantics, visual narratives, monsters

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\* **Authors' contributions:** Philippe Schlenker initiated this research and developed the analyses. All examples were constructed by and with Jonathan Lamberton, and the main generalizations were formulated and tested jointly by the authors. In the last stage, Jonathan Lamberton met with Jason Lamberton to informally collect the latter's judgments.

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**Supplementary materials:** ASL and gesture videos, raw ASL data, and average computations can be found in this OSF folder: <https://osf.io/pm7es/overview>

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# 1 Introduction

## 1.1 Goals

It has long been accepted that human language exists in (at least) two modalities, the vocal and the visual one (e.g. Sandler and Lillo-Martin 2006). But the relation between spoken language on its own ("speech"<sup>1</sup>) and sign language ("sign") is not one of mere structural identity. One reason is that sign can make rich use of iconic operations that have no direct counterpart in speech—although they exist in gestures that accompany speech. On a methodological level, this led Goldin-Meadow and Brentari (2017) to make the influential injunction that sign with iconicity should be compared to speech with gestures rather than to speech alone. On a formal level, this led to the claim that natural language semantics should be analyzed not just within standard logical semantics, but rather within an extension thereof, "Iconological Semantics", which includes an explicit pictorial component (Schlenker and Lamberton 2024). In line with Goldin-Meadow and Brentari's injunction, it was proposed that this framework should also be applied to speech with gestures.

This piece is concerned with context-dependency and iconicity in sign, with a focus on ASL (American Sign Language). While the traditional view of indexicals was that their context of evaluation is rigidly fixed throughout a sentence (Kaplan 1989), cross-linguistic research of the last 25 years has found numerous cases in which the context can be shifted by attitude operators, which are 'monsters' in Kaplan's terminology (e.g. Schlenker 1999, 2003; Anand and Nevins 2004; Anand 2006; Deal 2020).<sup>2</sup> In spoken languages, context shift is covert, but in sign languages it has been argued to be sometimes realized by Role Shift (Quer 2005, 2013, Schlenker 2017a,b), an operation in which the signer visibly adopts somebody else's perspective. It would be tempting to take Role Shift to just be overt context shift. But in ASL and LSF (French Sign Language), Role Shift involves peculiarities that have motivated the development of non-standard context-shifting operators ("Super Monsters"), and also of alternative treatments. First, Role Shift arises not just in attitude reports ("Attitude Role Shift", sometimes called "Constructed Dialogue") but also in action reports ('Action Role Shift', often called 'Constructed Action')<sup>3</sup>, where no monsters are found in spoken language. Second, Role Shift forces elements within its scope to be interpreted in a maximally iconic fashion ("Maximize Iconicity"), something that has not been found in spoken language monsters (Schlenker 2017b).

A further specificity of sign (shared with gestures) pertains to visual iconicity, which can only be evaluated by reference to a viewpoint or "camera position" (see for instance Greenberg 2013, 2021 and Abusch 2013, 2020). In Iconological Semantics, viewpoint manipulation is the 'glue' that connects iconic (pictorial) semantics with standard logical semantics in sign language. Specifically, the object language is endowed with viewpoint variables, which may be free and have a value provided by the context. While this value may correspond to the signer's current viewpoint, in many cases this is not so—viewpoint variables may denote different salient viewpoints besides the signer's. And in some cases, viewpoint variables may depend on quantifiers, which gives rise to a complex interaction between logical and iconic resources.

We revisit Role Shift from the perspective of Iconological Semantics, and argue that a key property, familiar from informal descriptions, has been missed by formal accounts: Role Shift usually triggers viewpoint shift. Our main argument is that under Action Role Shift, classifier predicates, whose position and movement are free and iconically interpreted, are normally evaluated with respect to a shifted viewpoint. This viewpoint shift derives Maximize Iconicity because all iconic elements that appear under Role Shift must by default be evaluated relative to the shifted viewpoint. While our main data are from ASL, we illustrate the main intuitions with gestural versions of Role Shift, which appear to display related generalizations (a full investigation is left for future research).

<sup>1</sup> We use terms such as "speech act" with a broad meaning encompassing sign.

<sup>2</sup> Kaplan 1989 (p. 510) emphasized that monsters, i.e. operators that shift the context of evaluation of indexicals, are logically well-formed but do not exist in natural language (an incorrect claim, as we saw).

<sup>3</sup> See Metzger 1995 for the terminology in terms of Constructed Action and Constructed Dialogue. While the terms (Action Role Shift vs. Constructed Action, Attitude Role Shift vs. Constructed Dialogue) come from different theoretical traditions, we use them descriptively here.

We develop an initial account in which Role Shift is just overt viewpoint shift. But this account arguably misses some important facts, and we thus develop an alternative in which Role Shift involves context shift with viewpoint shift ("context+viewpoint shift"). This is natural because the signer plays a dual role: as the center of the context, but also as the "puppeteer" with respect to whom highly iconic elements are initially evaluated.<sup>4</sup> As a result, when the signer shifts positions, it is natural that both the context and the viewpoint should be shifted as well. We do not aim to fully adjudicate between the two accounts, as each has its strengths and weaknesses. In addition, some key datapoints are subtle and will have to be investigated more systematically.

While our formal analyses are new, the driving intuition isn't. Gestural and sign language research alike distinguish between observer viewpoint, which helps depict a scene from a salient external viewpoint, and character viewpoint, which involves the viewpoint of a character within the scene (McNeill, 1992; Parrill, 2009, 2010). And it is certainly an old idea that Role Shift is associated with a character viewpoint depiction (Cormier et al. 2012, Quinto-Pozos and Parrill 2015; see also Janzen 2025). As we will see, cashing out this intuition has far-reaching consequences for the formal analysis of Role Shift and of Super Monsters.

## 1.2 Main findings

To get an intuitive idea of our main findings, an example might help. In (1), the context specifies that Ann was running in a straight line, displayed by the classifier movement in (2)a. The sentence in (1)a involves no Role Shift, and iconically displays the movement of an elephant, as illustrated in (2)b. (1)b is identical to (1)a, with the difference that the signer has rotated, signaling a clear Role Shift. Without Role Shift, the elephant movement can be interpreted relative to the original viewpoint, and thus as coming from Ann's front, slightly to the right. With Role Shift, for our main consultant, only a relative interpretation is possible: Ann's viewpoint is adopted and the elephant is understood to come, more or less orthogonally, from her right, matching the movement of the elephant classifier relative to the signer's shifted orientation in (2)b.

- (1) Context (see (25) for the full example):

'Yesterday, Ann's eyes were itching. She was at her circus. Further away was my zoo, which has no fence. Ann was running towards my zoo.

a. JOG PALMS-UP EYES-DROP, SCARED, ELEPHANT <sup>scared eyes</sup> a-ANIMAL-b-cl.  
 [Ambiguous] She stopped running to put drops in her eyes and she got scared by an elephant **coming (i) from the front (on the right) [3/3 judgments] or (ii) from the rear right side [2/3 judgments].**

RS<sub>b</sub> \_\_\_\_\_

b. JOG PALMS-UP EYES-DROP, SCARED, ELEPHANT <sup>scared eyes</sup> a-ANIMAL-cl-b.  
 [Unambiguous] She stopped running to put drops in her eyes and she got scared by an elephant **coming from the right [3/3 judgments].**

(ASL 37, 3480, <https://youtu.be/DXOoLe2j3wI>)

<sup>4</sup> We refer to the signer as 'puppeteer' when viewpoint is essential, as it is with reference to this puppeteer that viewpoint is defined in Iconological Semantics.

## (2) Loci and classifier movement in (1)

## a. Movement of the classifier representing Ann in the context

Ann classifier movement



## b. No Role Shift in (1)a (no signer rotation):

absolute/relative interpretation of elephant movement

elephant classifier movement



## c. Role Shift in (1)b (signer rotation):

only relative interpretation of elephant movement

elephant classifier movement



Thus Action Role Shift forces the iconic viewpoint to be shifted to that of the individual that triggers Role Shift. There are several complications, including that without Role Shift the viewpoint may be shifted too—what is special about Role Shift is that viewpoint shift is obligatory, as stated in (3). A further complication is that in Attitude Role Shift, it is the entire role-shifted clause that is iconically reproduced from the shifted viewpoint (e.g. Schlenker 2017a,b, Davidson 2015).

## (3) Viewpoint Shift Generalization

While viewpoint shift is optional in the absence of Role Shift, with Role Shift it is usually obligatory: the iconic viewpoint must correspond to that of the individual that triggers Role Shift. Specifically:

a. under Action Role Shift, classifiers must be evaluated relative to the shifted viewpoint;

b. under Attitude Role Shift, the entire role-shifted clause must pictorially (quotationally) reflect a signing event seen from the attitude holder's viewpoint.

## 1.3 Structure

After describing our elicitation methods and transcription conventions, we provide some background on viewpoint shift and Role Shift in Section 2. In Section 3, we investigate Action Role Shift and establish that ASL Role Shift usually implies viewpoint shift; the main intuition is also illustrated with gestural data. In Section 4, we consider Attitude Role Shift and argue that the viewpoint of evaluation of individual classifiers is fairly free, but that the entire role-shifted clause must iconically depict a signing event seen from the attitude holder's viewpoint. In Section 5, we develop a minimal theory in which Role Shift is just overt viewpoint shift. After highlighting some of its potential shortcomings, we develop in Section 6 an alternative that takes Role Shift to involve context shift with viewpoint shift (= context+viewpoint shift). In line with old ideas about the connection between Role Shift and gestural enactment, we discuss the import of gestural counterparts of Role Shift in Section 7, and we draw some conclusions in Section 8.

## 1.4 Elicitation methods and transcription conventions

The main consultant (and co-author) is a Deaf, native signer of ASL (of Deaf, signing parents).<sup>5</sup> Our goal is to describe his judgments in a transparent fashion; we thus describe our main consultant's idiolect. Elicitation was conducted using the playback method, described for instance in Schlenker et al. 2013, Schlenker and Lamberton 2019, 2022. It involved repeated quantitative acceptability judgments (on a 7-point scale with 7 = best), as well as inferential judgments. Average scores appear as superscripts at the beginning of transcriptions. Raw scores appear in parentheses (e.g. (5, 7, 3)) when there was more than a 2-point difference on the same question across different sessions. When

<sup>5</sup> We use the term *consultant* to refer to a collaborator that assesses sentences, including if this person is also a contributor to the article.

inferences are described, *1/3 judgment* means that the inference was obtained in 1 judgment out of 3; similarly for *2/3* and *3/3 judgments* (these may sum to more than 3 in case several inferences were concurrently accepted). References such as (e.g. *ASL 37, 3334, 3 judgments*) at the end of paradigms cross-reference the ASL video (here video 37, 3334) and indicate the number of iterated judgment tasks (on different days, here three). We systematically provide links to anonymized versions of the source ASL videos (when new), and specialists are invited to consult the raw judgments in the Supplementary Materials B when relevant. After the data were collected, Jonathan Lamberton informally obtained the judgments of his brother Jason Lamberton, who is also a Deaf native signer of ASL. Some but not all contrasts were replicated. The main points of disagreements are noted as we go, and summarized in Sections 3.5 and 4.3. A full description appears in the Supplementary Materials C. Unless noted otherwise, the two consultants agreed with respect to the contrasts we report.

Transcription conventions are standard for sign language. Loci are alphabetized from dominant to non-dominant side from the signer's perspective (here: from right to left). A suffixed locus, as in *WORD-i*, indicates that the word points toward locus *i* (a position of signing space associated with a discourse referent). *IX-i* (for 'index') is an index sign pointing toward locus *i*. *EXPRESSION<sub>i</sub>* is used for some words associated with locus *i* by virtue of being signed in (rather than by pointing to) the corresponding area of signing space. Agreement verbs include loci in their realization—for instance, the verb *a-HIT-b* starts out from locus *a* and targets locus *b*. We put *-cl* at the end of classifier predicates, and describe their movement with affixed words (e.g. *TREE-moving on the signer's left-cl*). For clarity, we sometimes append a picture of the classifier right after its transcription. We do not transcribe non-manuals except when relevant to Role Shift; readers are invited to consult the anonymized videos for further details of realization. Role Shift is transcribed as *RS*\_\_\_\_, with a line indicating its extent. Other non-manuals are also transcribed with a line, with ad hoc notations—e.g. *excited*\_\_\_\_ for an excited facial expression co-occurring with the words below the line.

## 2 Background on Viewpoint Shift and Role Shift

### 2.1 Viewpoint shift in ASL

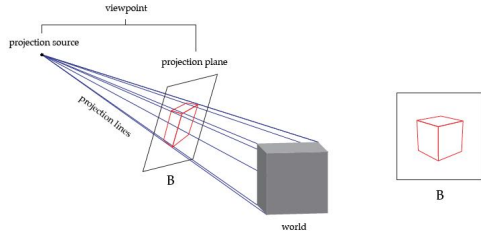
Sign languages have diverse constructions with an iconic component. The clearest case involves classifier predicates (classifiers for short). They have a lexically specified form, but their position in signing space, their orientation and their movement (if there is one) are unconstrained, and interpreted iconically. Visual iconicity requires a viewpoint relative to which it is evaluated.

Schlenker and Lamberton 2024 propose a formal framework that combines standard logical semantics with pictorial semantics in the tradition of Greenberg 2013, 2021 and Abusch 2013, 2020. The Greenberg/Abusch pictorial semantics is defined in terms of static 2D geometric projections, but it can be adapted to the case of 3D animations used in sign language. Schlenker and Lamberton's proposal builds on the fact that pictorial semantics provides a definition of truth for pictures relative to a time, a world, an assignment function and a viewpoint. In brief, a picture is true at a time *t* in a world *w* if at *t* the world *w* projects to the picture in view of relevant projection method.<sup>6</sup> For example, in (4), a cube projects onto the picture (the projection plane) according to the rules of perspective projection. But this requires the specification of a viewpoint, made in this case of a projection source and of the projection plane.

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<sup>6</sup> Here and below, we follow the summary of Schlenker et al. 2025.

(4) **An example of a projection method: perspective projection (Greenberg 2021)**



To this framework with viewpoints, times and worlds, Abusch 2013, 2020 adds a dependency on assignment functions. Her motivation is that in silent pictorial narratives, there are ambiguities of coreference that are elegantly handled by positing covert variables, and hence assignment functions.

Three of the parameters we just mentioned, namely time, world, and assignment function, are standard in logical semantics. Therefore only viewpoint-dependency requires special measures to allow for a unification of logical and pictorial semantics. In Iconological Semantics, classifiers (and iconic expressions more generally) come with a viewpoint variable (e.g.  $\pi$ ), which simultaneously tells the semantic system that the expression must be interpreted iconically, and specifies the viewpoint relative to which pictorial interpretation proceeds. To illustrate, we can start from the lexical entry for a standard nominal (not a classifier), *PLANE*, in (5)a. The minimally different airplane classifier in (5)b, *PLANE-cl*, has this lexical component as well, but in addition it has a projective component (boldfaced).

- (5) a.  $[[\text{PLANE}]]^{c, s, t, w} = \lambda x_e . \text{plane}'_{t, w}(x)$   
 b.  $[[\text{PLANE-cl}]]^{c, s, t, w} = \lambda x_e . \text{plane}'_{t, w}(x) = 1$  and  **$\text{proj}(x, s(\pi), t, w) = \text{plane-cl}$**

In essence, an object  $x$  is in the extension of *PLANE-cl* <sup>$\pi$</sup>  relative to assignment function  $s$  at time  $t$  in world  $w$  just in case, at  $t$  and in  $w$ ,  $x$  is a plane, and in addition  $x$  projects to the position and orientation specified by the classifier relative to viewpoint  $s(\pi)$ . A notion of dynamic projection is further defined in Schlenker and Lamberton 2024, aimed at capturing the case in which classifiers move and are thus animations rather than static representations. We will make use of both static and dynamic projection below, without getting into the details of how the notions should be defined because our focus is on context- and viewpoint-dependency, not on the iconic system per se. The general rule of interpretation we will employ for classifiers is given in (6). It follows all earlier accounts we know of in taking the iconic contributions of expressions that have their own time slot to be (potentially) at-issue.<sup>7</sup>

- (6) If  $P$  is a 1-place predicate and if  $\pi$  is a viewpoint-denoting expression  
 $[[P]]^{c, s, t, w} = \lambda x_e . [[P]]^{c, s, t, w}(x) = 1$  and  **$\text{proj}(x, [[\pi]]^{c, s, t, w}, t, w) = P$**

In simple cases, there is a single viewpoint variable whose value is provided by the context. This value may but need not be the signer's current viewpoint. In more complex cases, a viewpoint variable may be bound by an existential quantifier in the scope of other operators, and/or several viewpoint variables may appear in the same clause. For instance, in (7)a, with the Logical Form in (7)b, a person classifier displaying a person walking, *person-walk-cl* <sup>$\pi$</sup> , is evaluated relative to the viewpoint variable  $\pi$ , which is bound by an existential quantifier (boldfaced). The resulting meaning is that 'always, there is some salient viewpoint relative to which a student walked as displayed by the classifier'.

<sup>7</sup> Zucchi 2011, 2017, Davidson 2015 and Schlenker 2017a treat the iconic component of classifiers and other expressions as conjoined with the lexical component. Schlenker 2018b develops a general theory of "iconic pragmatics" according to which iconic expressions that have their own time slot, in particular pro-speech (= speech-replacing) gestures and sign language classifiers, have an at-issue component, whereas co-speech and co-sign gestures trigger conditionalized presuppositions (called "cosuppositions"). In line with findings about pro-speech gestures, classifiers may, in addition, make various other contribution types (Schlenker 2019, Tieu et al. 2019, Schlenker et al. 2025).

The present framework includes two small differences relative to that of Schlenker and Lamberton 2024. First, we abstract away from issues of trivalence. Second, we allow for viewpoint expressions that are not viewpoint variables. This will come in handy in Section 5.4, where we countenance functional viewpoint expressions.

- (7) *Context*: This school has 4 classrooms, one for each of 4 teachers (each teacher always teaches in the same classroom).  
 a. <sup>7</sup> CLASS BREAK ALL ALWAYS HAVE STUDENT PERSON-walk-back\_right-cl.  
 'In all classes, during the break, there is always a student that leaves toward the back, to the right.'  
 b. always<sub>T</sub>  $\exists \pi$  there-is student person-walk-cl<sup>7</sup>  
 (ASL, 35, 2254b; 3 judgments; <https://youtu.be/h7EnRK6poAE>)

Although Schlenker and Lamberton 2024 take viewpoints to be static, in more complex examples, discussed in Schlenker et al., to appear, viewpoints may move, giving rise to sign language counterparts of "traveling cameras" (as argued for instance in Liddell 2003 and Müller 2018).

The Greenberg/Abusch pictorial semantics is defined in terms of 2D geometric projections, but it must be adapted to the case of 3D representations used in sign language. As a result, the notion of a viewpoint is modified as well. In a nutshell, for Schlenker and Lamberton 2024, the viewpoint is the real-world observation point from which a scene is observed and then recreated by the signer in a scaled version by way of puppet-like elements (classifiers), as illustrated in (8). Schlenker and Lamberton take the viewpoint to include a Cartesian coordinate system, relative to which the position and orientation of real-world objects can be specified.<sup>8</sup>

- (8) **Obama sitting across from the Dalai Lama: from the scene to its classifier representation**<sup>9</sup>  
 a. Original scene, with the viewpoint (☺) the Dalai Lama  
 b. The signer recreates a scaled version of the 3D scene with two classifiers representing Obama and the Dalai Lama



It is important to note that the examples discussed in Schlenker and Lamberton 2024 do not include Role Shift, a construction to which we now turn.

## 2.2 Role Shift in ASL

### 2.2.1 Main facts

As summarized in Quer 2013, Role Shift across sign languages is morpho-syntactically characterized by non-manual markers such as the following: (i) "temporary interruption of eye contact with the actual interlocutor and direction change of eye gaze towards the reported interlocutor"; (ii) "slight shift of the upper body in the direction of the locus associated with the author of the reported utterance"; (iii) "change in head position"; (iv) "facial expression associated to the reported agent".

Role Shift in ASL comes in two varieties (e.g. Metzger 1995, Schlenker 2017a). Attitude Role Shift (= Constructed Discourse) serves to report speech or thoughts, sometimes (but not always) with an explicit verb of indirect discourse. While it is sometimes analyzed as a quotational marker, it has been argued to have properties (such as extraction of interrogative elements out of the role-shifted clause) which are not possible in English-style quotations (but see arguments for the opposite conclusion in Lee et al. 1997). If it is analyzed as indirect discourse, it involves context shift, as indexicals in ASL Attitude Role Shift must be interpreted relative to reported context. In addition, Attitude Role Shift has a quotational component.<sup>10</sup>

<sup>8</sup> As Schlenker and Lamberton 2024 discuss, one may define things either in terms of the "puppeteer's" viewpoint, or in terms of the audience's viewpoint. We will talk in terms of the puppeteer's viewpoint below.

<sup>9</sup> From Schlenker and Lamberton 2024. Picture credit (Obama and the Dalai Lama): Pete Souza, Public domain, via Wikimedia Commons (we use a mirror-image version).

<sup>10</sup> Koulidobrova and Davidson 2015 introduce a further ramification among two classes of ASL predicates that are compatible with Attitude Role Shift, namely *think*-type verbs ("doxastic verbs") and *say*-type verbs ("proffering verbs"). They claim that under ASL *think*-type but not *say*-type verbs, (i) Role Shift can optionally affect the verb (not just the complement clause), and (ii) the first person pronoun can optionally refer to the actual



Action Role Shift (= Constructed Action) serves to report actions in a particularly vivid fashion. It often targets a single verb, which can take first person agreement markers, but in earlier descriptions (Schlenker 2017a), full indexicals cannot appear in the scope of Role Shift. Action Role Shift too has a strong iconic component: iconically interpretable material must be attributed to the agent, not to the signer.

Attitude Role Shift and Action Role Shift are illustrated in (9)a and (10)a respectively. (9)b and (10)b are deviant because an iconic element, namely a happy face on the signer's part [transcribed as :-)] , starts before Role Shift—indicating that is evaluated with respect to the signer—and continues under Role Shift. This yields deviance because under Role Shift, iconic elements must be evaluated with respect to the agent of the reported speech or thought act (without Role Shift, no such problem arises).

- (9) SEE THAT-a ARROGANT FRENCH SWIMMER IX-a? YESTERDAY IX-a ANGRY.

'See that arrogant French swimmer? Yesterday he was angry.'

a. RS<sub>a</sub>\_\_\_\_\_

<sup>6.2</sup> IX-a SAY IX-1 WILL LEAVE.

He said: 'I will leave.'

b. :-)-----

RS<sub>a</sub>\_\_\_\_\_

<sup>4.6</sup> IX-a SAY IX-1 WILL LEAVE.

*Rating under the meaning:* the SPEAKER is displaying his happiness that the French swimmer said he was leaving

He said: 'I will leave.'

(ASL, 14, 233; Schlenker 2017b, 2018a)

- (10) SEE THAT ARROGANT FRENCH SWIMMER IX-a? YESTERDAY IX-a ANGRY.

a. RS<sub>a</sub>\_\_\_\_\_

<sup>7</sup> IX-a 1-WALK-WITH-ENERGY(CL-ONE).

He left with energy.'

b. :-)-----

RS<sub>a</sub>\_\_\_\_\_

<sup>3.6</sup> IX-a 1-WALK-WITH-ENERGY(CL-ONE).

*Rating under the meaning:* the SPEAKER is displaying his happiness that the French swimmer was leaving

He left with energy.'

(ASL, 14, 233; Schlenker 2017b, 2018a)

### 2.2.2 Theoretical directions

How should Role Shift be analyzed? In various descriptions, it has been linked to strategies of enactment in gestures (e.g. Liddell and Metzger 1998, Liddell 2003, Lillo-Martin 2012, Quinto-Pozos and Parrill 2015; see also Emmorey et al. 2000), and more specifically to character viewpoint gestures, in which a scene is described from the viewpoint of a character within it (Cormier et al. 2012, Quinto-Pozos and Parrill 2015, Waller 2021; McNeill, 1992; Parrill, 2009, 2010); these contrast with observer viewpoint gestures, in which a scene is depicted from a salient external viewpoint.

Within the formal literature, there have been two broad analytical directions. Quer 2005, 2013 and Schlenker 2017a,b have argued that Attitude Role Shift is an overt instance of context shift because some or all indexicals that appear in its scope obtain a shifted interpretation (for theories of context shift, see for instance Schlenker 1999, 2003, Anand 2006, Deal 2020; see also Maier 2018 and Maier and Steinbach 2022 for alternatives). Schlenker 2017a,b proposes an implementation and argues that expressions that appear under Role Shift should be interpreted *maximally iconically*, i.e. so as to maximize the possibilities of iconic projection between the signs used and the situations they make

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speaker (rather than just to the agent of the reported attitude). Our examples of Attitude Role Shift in this piece involve the *say*-class; we leave an investigation of viewpoint shift in relation to the *think*-class for future research.

reference to. Following many others (e.g. Clark and Gerrig 1990), Schlenker 2017b further argues that quotation can be seen as a special and particular stringent case of iconicity, and that the condition of Maximal Iconicity can thus capture properties of both Attitude and Action Role Shift. A key problem with this analysis is that maximal iconicity is in essence a stipulation: it doesn't follow from anything else in the theory. The earlier intuition that enactment is involved is captured indirectly: because of context shift, the signer represents another individual (the agent of the shifted context); and because of Maximal Iconicity, iconic aspects of the signs are interpreted relative to that shifted context.

Following Lillo-Martin 1995, 2012, Davidson 2015 criticized the analysis of Schlenker 2017a,b<sup>11</sup>, and took Role Shift to indicate that the role-shifted signs have a demonstrative component, and are in effect both signs and co-sign gestures (see also Maier 2018). Specifically, Davidson follows Supalla 1982 in proposing that in Role Shift the signer's body behaves as a classifier and is used to demonstrate another person's signs, gestures or actions. But the thrust of the analysis is to connect Role Shift to a broader class of constructions involving event depictions. For Attitude Role Shift, Davidson adopts an analysis, inspired by Clark and Gerrig 1990, in which quotation is a special sort of demonstration. With *SAY*, the meaning obtained is in essence: "say like this". For Action Role Shift, she takes role-shifted verbs to be interpreted with a kind of "like this" modifier that refers to their own form. While it is not clear what it would mean for an entire VP or a larger structure to be interpreted demonstratively in this way, Davidson argues that Action Role Shift in fact applies to verbs alone (although there are some examples with larger structures in Schlenker 2017a,b). In this analysis, the intuition of enactment is captured more directly by the idea that the signer's body acts as a classifier, whose behavior is iconically interpreted.

### 2.3 Role Shift in gestures?

While our goal is to account for Role Shift in ASL, we will discuss gestural examples because they have two virtues: they help illustrate the main intuitions for non-signers, and they highlight that the key generalizations might not be restricted to sign languages. (These gestural data should be taken as illustrative at this point. They are based on our intuitions and those of a (non-signing) native speaker of American English, collected with informal methods, unlike our ASL data [see Section 1.4]; we leave a systematic and possibly experimental investigation for future research.)

The gestural connection is nothing new. From the start, descriptions of Role Shift (under the name of Constructed Action/Dialogue) hinted at a connection with strategies of reenactment in gestures (see for instance the early references cited in Lillo-Martin 2012 and Quinto-Pozos and Parrill 2015, including Metzger 1995 and Liddell and Metzger 1998). Analyses that took Role Shift to involve a body classifier (Supalla 1982, Davidson 2015) went in the same direction. Coming from a slightly different perspective, Schlenker 2020 listed several gestural constructions that might have a grammar reminiscent of sign language. This included Role Shift, both in its Action and in its Attitude versions, as illustrated in (11).

*Notation:* Gestures are transcribed in capital letters in a non-standard font. A co-speech gesture *G* co-occurring with an expression *E* is written as *G* [*E*] (with the modified expression boldfaced). *RS<sub>i</sub>* indicates that the speaker shifts his body to adopt the position of a fictional character found in gestural locus *i* (here we will have *i* = a or *i* = b). The gesture that follows *RS<sub>i</sub>* is realized from this shifted position.

- (11) <sup>7</sup> I was standing next to IX-hand-a [**little Robin**] and IX-hand-b [**little Francis**], and I was holding a really yummy chocolate bar. And I asked: Who wants it? And so of course
- (i) IX-hand-a [**little Robin**] goes: *RS<sub>a</sub>* IX-1. And IX-hand-b [**little Francis**] goes: *RS<sub>b</sub>* IX-1.
- Next thing I know,
- (ii) IX-hand-a [**little Robin**] turns to IX-hand-b [**Francis**] and [*RS<sub>a</sub>* SLAP-b]. And so IX-hand-b [**Francis**] *RS<sub>b</sub>* PUNCH-a.
- (Video 4053; <https://youtu.be/r0dhqgOk2k0>)

<sup>11</sup> Schlenker 2017a,b first appeared on LingBuzz in April 2014.

The first part of (11) (namely (i)) involves gestural counterpart of Attitude Role Shift, while the second part (i.e. (ii)) displays a version of Action Role Shift; let us call these Gestural Attitude Role Shift and Gestural Action Role Shift respectively. Their experimental acceptability was established (with different sentences) in Tieu et al. 2025.

The analogy with ASL Role Shift can be strengthened. As we argue in Appendix III, it is possible to replicate in gestures the observation that full indexicals cannot appear under ASL Action Role Shift but can under Attitude Role Shift (first person agreement markers are unconstrained, by contrast). More crucially for present purposes, we will see below that Gestural Action Role Shift might involve the same kind of viewpoint shift as ASL Role Shift as stated in (3)a (the quotational nature of Gestural Attitude Role Shift is apparent in (11)(i) above, but since it involves a single expression, namely the self-pointing index, it is less than striking).

## 2.4 Outlook

We will argue that both analytical lines in the formal tradition—the context-shifting analysis, and the demonstrative analysis—have missed an essential component of Role Shift, familiar from informal descriptions: Role Shift usually shifts the viewpoint of evaluation of iconic elements (Cormier et al. 2012, Quinto-Pozos and Parrill 2015, Waller 2021; McNeill, 1992; Parrill, 2009, 2010); and this generalization arguably extends to Gestural Role Shift.<sup>12</sup> To assess viewpoint shift, we will investigate systematic paradigms in which classifiers appear under Role Shift. Our main finding is that Role Shift, unlike classifiers on their own, triggers viewpoint shift by default, as encapsulated in the Viewpoint Shift Generalization in (3)a. This will cause a problem for "Role Shift as body classifier" theories, for two reasons. First, Role Shift crucially differs from standard classifiers in triggering viewpoint shift. Second, Action Role Shift can be applied to complex constructions, contrary to the claim that it can only affect a single verb.

We will explore two formal analyses of the phenomenon. In a minimal theory, Role Shift is treated as overt viewpoint shift, without context shift. This analysis encounters two potential difficulties, however. First, it does not account for the fact that Attitude Role Shift displays greater grammatical integration than standard quotation (e.g. Schlenker 2017a). Second, it does not account for the observation that indexicals are usually degraded under Action Role Shift (Schlenker 2017a, partly contradicted by Schlenker et al., to appear). To address these problems, we will develop a theory in which Role Shift involves both context and viewpoint shift. This can be seen as an improvement of the "Role Shift as overt context shift" analysis. It is conceptually natural because a context determines a viewpoint, corresponding to the perspective of the agent. Shifting the context shifts this viewpoint as well, hence the fact that Role Shift implies viewpoint shift.

Importantly, each theory has its own strengths and weaknesses, and thus we believe the debate will have to be refined and extended in future research. In addition, some key data points will have to be revisited.

## 3 Action Role Shift and Viewpoint Shift

Our goal in this section is to establish the generalization announced in (3)a: Action Role Shift entails viewpoint shift, and the shifted viewpoint is closely associated with the character whose perspective is adopted. By contrast, in the absence of Action Role Shift, viewpoint shift is optional and depends on contextual factors.

### 3.1 Gestural motivation

We start with gestural examples to illustrate the main idea, and to help non-signers grasp the main intuitions. The paradigm in (12) tests whether Gestural Action Role Shift implies viewpoint shift, with

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<sup>12</sup> Waller 2021 (p. 56) distinguishes between vantage-point shift and character role shift. The former can involve eyegaze shift and torso shift but not "enactment of facial expression or posture of the character". By contrast, "character role shift requires a character to inhabit", and involves enactment. We come back to this point in the conclusion.

positions sketched in (13). When the speaker doesn't rotate, as in (12)a,b, Ann's movement can be represented with an observer perspective (as in (12)a) or with a character perspective ((12)b). But when the speaker realizes gestural version of Role Shift (with body shift and eyegaze shift), it seems to us difficult to keep an observer viewpoint (we take (12)c to be degraded, which we notate as <#>).

(12) **Horizontal dimension only**

Tomorrow Ann will be IX-a [at the office nearby], and there will be an alarm drill. She doesn't take these things very seriously, so when the alarm goes off, she will

a. No Gestural Action Role Shift - No viewpoint shift

leisurely \_\_\_\_\_  
a-WALK-b.

<https://youtu.be/2QigyU0j9wA>

b. No Gestural Action Role Shift - Viewpoint shift

leisurely \_\_\_\_\_  
I-WALK.

<https://youtu.be/S4V5T5pDYa8>

c. Gestural Action Role Shift - No viewpoint shift

<#> <sup>leisurely</sup> RS<sub>a</sub> a-WALK-b.

<https://youtu.be/xj0a0wV0JZg>

d. Gestural Action Role Shift - Viewpoint shift

leisurely \_\_\_\_\_  
RS<sub>a</sub> I-WALK.

<https://youtu.be/rZtWPdtwesI>

(13) Finger movement representing Ann walking in (12)

a. Example (12)a



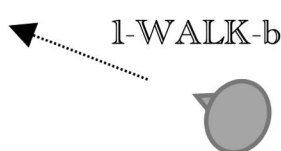
b. Examples (12)b



c. Example(12)c



d. Examples (12)d



Importantly, the shifted viewpoint in (12)d, with the gesture illustrated in (13)d, cannot be literally the character's perspective. The reason is that Ann is represented by fingers moving in front of the speaker, and this couldn't represent the scene as perceived by Ann. Rather, the scene is represented from a viewpoint associated with Ann—say right behind Ann's initial position. A similar remark will apply to our sign language examples.

While only the horizontal dimension is involved in (12), the vertical dimension plays a role in (14), with positions represented in (15). To us, (14)c is degraded—under Gestural Action Role Shift, it is difficult to maintain the original (observer) viewpoint, and a viewpoint associated with the character must be adopted instead.

(14) **Vertical dimension**

At 1pm, Ann will be standing next to the exit door, ready to skydive. And then she will

a. No Gestural Action Role Shift - No viewpoint shift

a-JUMP-down.

<https://youtu.be/tBIBA-e3Qn4>

b. No Gestural Action Role Shift - Viewpoint shift

1-JUMP-down.

[https://youtu.be/JVg-0de\\_yj8](https://youtu.be/JVg-0de_yj8)

c. Gestural Action Role Shift - No viewpoint shift

<#> RS<sub>a</sub> a-JUMP-down.

<https://youtu.be/lsAxZYiNQmI>

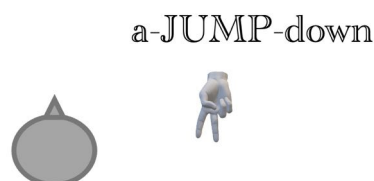
d. Gestural Action Role Shift - viewpoint shift

RS<sub>a</sub> 1-JUMP-down.

<https://youtu.be/q64FAJ5Xxs>

(15) Fingers representing Ann jumping in (14)

a. Example (14)a

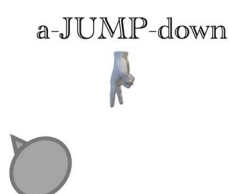


b. Example (14)b

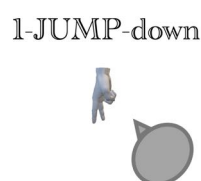
1-JUMP-down



c. Example (14)c



d. Examples (14)d



Finally, in (16) we explore a complex construction to determine whether a more deeply embedded gesture produced under Gestural Action Role Shift is preferably interpreted with viewpoint shift. Ann's plane is represented as moving from *a* to *b* (see the loci in (17)). Then (i) Ann is depicted as piloting, and (ii) another plane movement is represented. In (16)a, (i) and (ii) are produced without Gestural Role Shift, and we expect that the other plane's movement can be interpreted without viewpoint shift (although this need not be the only option). In (16)b, Gestural Role Shift is applied, and the other plane's movement is interpreted from Ann's perspective.

(16) **Complex constructions**

During her flight test tomorrow, Ann will PLANE<sub>a\_right\_hand</sub> [start from Base A] and a-FLY-b<sub>right\_hand</sub> [fly to Base B]. But there will be PLANE<sub>neutral\_left\_hand</sub> [another plane flying at the same time]. So it's possible that as Ann

a. PILOT-INTENSELY, b-FLY-a.

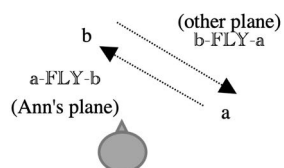
<https://youtu.be/-1NwrhfKjvE>

b. RS PILOT-INTENSELY, b-FLY-a.

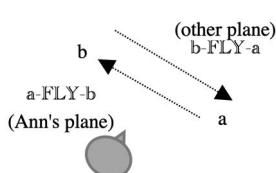
Expected interpretation: as Ann pilots, a plane will cross her path in front of her from her left to her right

<https://youtu.be/ZetFWKNGS9Y>

(17) a. No Gestural Role Shift



b. Gestural Role Shift



In sum, these preliminary gestural examples suggest that iconic construction that appear under Gestural Action Role Shift are preferably interpreted with viewpoint shift. We will now see that similar findings hold in ASL.

### 3.2 *Action Role Shift triggers viewpoint shift: bare classifiers in the horizontal dimension*

The telltale sign of viewpoint shift in ASL is the interpretation of classifiers, which are the clearest viewpoint-sensitive constructions. In line with our gestural examples, we will establish our generalization on the basis of three construction types, systematically investigated with and without Action Role Shift: bare classifiers with a horizontal movement; bare classifiers that also involve a vertical movement; and complex constructions with a horizontal movement. Bare embedded classifiers are simplest to investigate, and they bear out the generalization in (3)a, but they are open to an alternative analysis in terms of agreement rather than viewpoint shift. Complex embedded constructions are harder to investigate but rule out this possibility.

As in our gestural examples in (12) and (14), the shifted viewpoint needn't literally be that of the relevant character, although it must be closely associated with that character—it may for instance be behind the character's original position, with the result that the character is represented as moving away from the viewpoint.

One issue we do not further pursue is the observation that Action Role Shift must always be somehow justified. A common intuition is that it must be warranted by an iconic component (see for instance Davidson 2015, Schlenker 2017a,b and Waller 2021<sup>13</sup>). Owing to our explicit treatment of viewpoint shift, our analysis has the means to assess such iconic effects; but determining what is "iconic enough" to license Action Role Shift is a topic we leave for future research.

We start by investigating bare classifiers with a horizontal movement, as illustrated in (18), with the classifier movements represented in (19)<sup>14</sup>, which should be reminiscent of the gestural example in (12).

(18) YESTERDAY ANN-a THERE-b COLLEGE. HAPPEN BOMB WARN. SO IX-a

'Yesterday Ann was at the college. There was a bomb warning. So she

excitedly\_\_\_\_\_

a.<sup>7</sup> b-PERSON-c-cl\_\_\_\_\_.

excitedly\_\_\_\_\_

b.<sup>6,7</sup> 1-PERSON-c-cl\_\_\_\_\_.

excitedly\_\_\_\_\_

c.<sup>7</sup> 1-PERSON-c-cl<sub>forward\_body\_movement</sub>.



<sup>13</sup> Waller 2021 writes (p. 58): "there is some evidence to support the notion that use of role shift is constrained to highly iconic signs. Among plain lexical verbs, there is slight evidence suggesting more iconic verbs are also more likely to appear under role shift. Among classifier constructions, the most iconic ones, handling classifier constructions, are on average the most likely to appear with elements of role shift."

<sup>14</sup> Here our second consultant agreed with the main contrasts but found the Role Shift unnecessary.

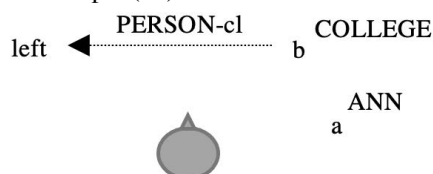
- RS<sub>a</sub> \_\_\_\_\_  
excitedly \_\_\_\_\_
- d. <sup>5</sup> b-PERSON-c-cl\_   
RS<sub>a</sub> \_\_\_\_\_  
excitedly \_\_\_\_\_
- e. <sup>6.3</sup> 1-PERSON-c-cl\_   
RS<sub>a</sub> \_\_\_\_\_  
excitedly \_\_\_\_\_
- f. <sup>7</sup> 1-PERSON-c-cl<sub>forward\_body\_movement</sub> \_\_\_\_\_

left running.'

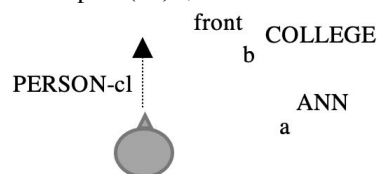
(ASL 37,3324, 3 judgments; <https://youtu.be/wUWYFxCMFHg>)

(19) Loci and classifier movement in (18)

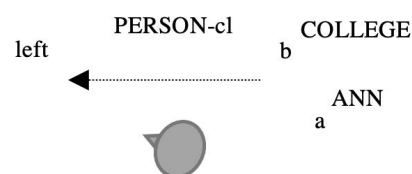
a. Example (18)a



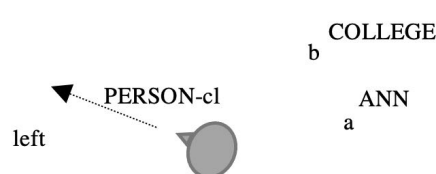
b. Examples (18)b,c



c. Example (18)d



d. Examples (18)e,f



In (18)a, the classifier representing Ann's movement starts out in the college locus *b*, and no viewpoint shift is necessary between the introduction of *b* and the classifier appearance. In (18)b,c, by contrast, the classifier starts out from the signer's position. This means that the viewpoint is presented as being right behind Ann's initial position (this is one of the many cases in which the shifted viewpoint is not the character's viewpoint, but a viewpoint initially associated with the character). There is a difference between (18)b and (18)c: in the former case, the signer's body is still, whereas in the latter, it moves forward to follow the classifier. The viewpoint is thus behind Ann and at a standstill in the first case, behind Ann and following her in the second.

The crucial example appears in (18)d. An attempt is made to apply Action Role Shift without shifting the viewpoint relative to which classifier movement is evaluated. Specifically, the signer rotates to adopt Ann's position, but the classifier movement remains as it was in (18)a, without Role Shift. The result is somewhat deviant, indicating that Action Role Shift is preferably accompanied with viewpoint shift. No such deviance is found in (18)e,f, where both Role Shift and viewpoint shift are applied, with the signer's body following the classifier in the second case but not in the first. Here too, the viewpoint is not Ann's, but one associated with Ann, since Ann's movement is depicted from that viewpoint.

A structurally similar example with a different context, temporal specification and classifier realization appears in (20). The signer predicts that Ann won't take the fire alarm drill very seriously, and will leave and go home at a leisurely pace. Here too, the example in (20)d is degraded, as Role Shift is not accompanied with viewpoint shift.

- (20) TOMORROW ANN<sub>a</sub> THERE<sub>b</sub> COLLEGE. POSS<sub>a</sub> HOME NEAR IX<sub>c</sub>. IX<sub>b</sub> WILL FIRE WARN  
'Tomorrow Ann will be at the college. Her home is nearby. There will be a fire alarm  
PRACTICE. IX-1 NOT THERE. IX<sub>a</sub> WILL  
drill. I won't be there, but she will

leisurely\_\_\_\_\_



- a. <sup>7</sup> b-PERSON-c-cl\_\_\_\_\_.

leisurely\_\_\_\_\_



- b. <sup>6,7</sup> 1-PERSON-c-cl\_\_\_\_\_.

leisurely\_\_\_\_\_

- c. <sup>6</sup> 1-PERSON-c-cl<sub>forward\_body\_movement</sub>•

RS<sub>a</sub>\_\_\_\_\_

leisurely\_\_\_\_\_



- d. <sup>4,7</sup> b-PERSON-c-cl\_\_\_\_\_.

RS<sub>a</sub>\_\_\_\_\_

leisurely\_\_\_\_\_



- e. <sup>7</sup> 1-PERSON-c-cl\_\_\_\_\_.

RS<sub>a</sub>\_\_\_\_\_

leisurely\_\_\_\_\_

- f. <sup>7</sup> 1-PERSON-c-cl<sub>forward\_body\_movement</sub>•

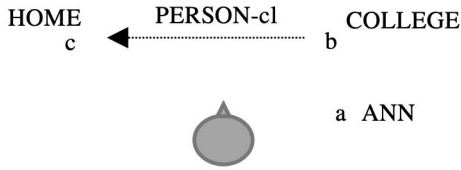
leisurely leave and go home.'

(ASL 37, 3346, 3 judgments; <https://youtu.be/Ou8GfUmrTbA>)

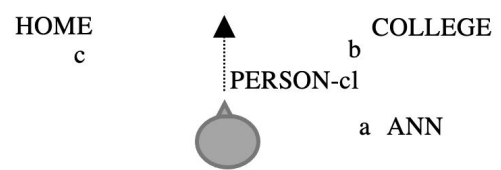


## (21) Loci and classifier movement in (20)

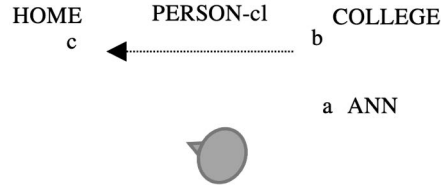
a. Example (20)a



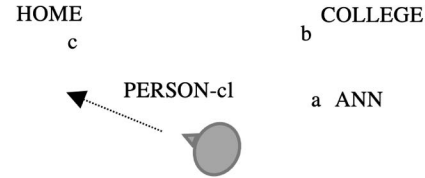
b. Examples (20)b,c



c. Example (20)d



d. Examples (20)e,f

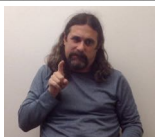


Three conclusions can be reached at this point. First, Action Role Shift seems to imply viewpoint shift. Second, there is a sharp contrast between Role Shift and classifiers, as it is clear that classifiers on their own (without Role Shift) do not require viewpoint shift; this will cause a difficulty for analyses of Role Shift as just involving a body classifier. Third, when Action Role Shift takes place, the signer's role-shifted position corresponds to the shifted viewpoint, which is also given by the agent's initial position. But this shifted viewpoint need not be identical to the agent's own position, as is made clear when a classifier representing the agent moves away from the signer's role-shifted position.

### 3.3 Action Role Shift triggers viewpoint shift: adding the vertical dimension

As we did in our motivational gestural examples in Section 3.1, we turn to cases in which viewpoint shift involves not just the horizontal but also the vertical dimension. In (22), Ann hesitates before jumping out of a plane to skydive. She is initially associated with locus *a*, on the signer's right. While the end of the sentence (*PALMS-moving-up*) is uniformly presented from Ann's viewpoint (with the ground nearing her), there are differences involving classifier movement and Role Shift. In (22)a, the jump is represented by a person classifier originating high in position *a*, and going down; the viewpoint is thus lower than Ann. In (22)b, the classifier originates at the level of the signer's body and moves down. The viewpoint of evaluation must thus be right behind Ann. (22)c,d are similar to (22)a,b, except that Role Shift is applied. (22)c is degraded—applying Role Shift makes it difficult to adopt a viewpoint not associated with Ann (it must be said, however, that the judgments were unstable). Importantly, in this case and in others pertaining to height, our second consultant was more liberal than our main consultant, accepting all sentences.

- (22) YESTERDAY ANN GO PARACHUTE. PLANE PLANE-cl<sub>circling a<sub>high</sub></sub> PLANE DOOR-OPEN.ANN<sub>a<sub>high</sub></sub>  
 'Yesterday Ann went parachuting. The plane circled, the door opened, and Ann

|                                                                                                                                                               |                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| hesitant_____<br><br>a. <sup>7</sup> PERSON-cl <sub>a<sub>high</sub></sub> | concerned_____<br><br>high-a-PERSON-falling-low-cl<br>PALMS-moving-up.<br>concerned_____ |
| hesitant_____<br><br>b. <sup>6</sup> PERSON <sub>high</sub> -cl            | <br>1-PERSON-falling-low-cl<br>PALMS-moving-up.                                          |

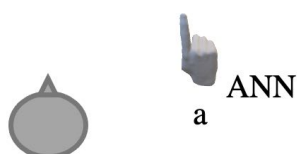
- RS<sub>a</sub> \_\_\_\_\_  
 hesitant \_\_\_\_\_ concerned \_\_\_\_\_
- c. <sup>5</sup> PERSON<sub>high-cl</sub> \_\_\_\_\_ **high-a**-PERSON-falling-low-cl PALMS-moving-up. (5, 7 3)  
 RS<sub>a</sub> \_\_\_\_\_  
 hesitant \_\_\_\_\_ concerned \_\_\_\_\_
- d. <sup>7</sup> PERSON<sub>high-cl</sub> \_\_\_\_\_ **1**-PERSON-falling-low-cl PALMS-moving-up.

hesitated, jumped, and panicked as the ground neared.'

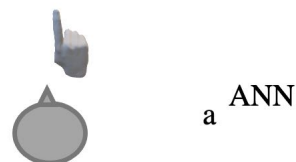
(ASL [37.3468](https://youtu.be/vlEWX37Et0Q), 3 judgments; <https://youtu.be/vlEWX37Et0Q>)

(23) Loci and classifier I on classifier position in (22)

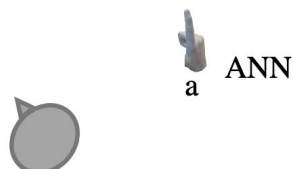
a. Example (22)a



b. Example (22)b



c. Example (22)c



d. Examples (22)d



The same generalization is supported by a structurally analogous paradigm, displayed in (24). It involves an enthusiastic rather than hesitant facial expression, but the locus configurations are similar to those depicted in (23). Here too, Role Shift implies viewpoint shift, which accounts for the deviance of (24)c (the differences in averages are sharp, but the judgments for (24)c were a bit unstable). (Our second consultant preferred not to have Role Shift here and found both role-shifted examples degraded.)

(24) YESTERDAY ANN INVOLVE DIVE COMPETITION. HAVE HIGH PLANK, PLANK-cl<sub>a</sub><sup>high</sup>. ANN  
 'Yesterday, Ann took part in a diving competition. There was a spring board high up. Ann

enthusiastic \_\_\_\_\_

a. <sup>6.7</sup> a<sup>high</sup>-PERSON-stands-then-jumps-low-cl \_\_\_\_\_

enthusiastic \_\_\_\_\_

b. <sup>6</sup> 1<sup>mid</sup>-PERSON-stands-then-jumps-low-cl \_\_\_\_\_

- RS<sub>a</sub>\_\_\_\_\_
- enthusiastic\_\_\_\_\_
- c. <sup>4</sup> a<sup>high</sup>-PERSON-stands-then-jumps-low-cl\_\_\_\_\_ (3, 6, 3)
- RS<sub>a</sub>\_\_\_\_\_
- enthusiastic\_\_\_\_\_
- d. <sup>7</sup> 1<sup>mid</sup>-PERSON-stands-then-jumps-low-cl\_\_\_\_\_



enthusiastically stood there and jumped.  
(ASL 37, 3452, 3 judgments; <https://youtu.be/JfD7hLx4fgk>)

In sum, for our main consultant, examples involving the horizontal dimension only or both the horizontal and the vertical dimension suggest that viewpoint shift is optional in the absence of Role Shift, but becomes preferred with Role Shift (see Appendix I-A for an apparent exception, which we suggest involves two successive Role Shifts)

There is an alternative interpretation of the data seen so far, however. One might think that it is because of agreement that the role-shifted classifier must start from the signer's role-shifted position: the subject is Ann, Role Shift is to Ann's position, and with agreement verbs it is usually assumed that subject agreement must target the signer's position in this case (e.g. Frederiksen and Mayberry 2018 Figure 3). Classifiers might fall under the same generalization. One might object that the gestural examples discussed in Section 3.1 display similar properties and yet could not involve agreement. But the latter point is not as obvious as it seems—Schlenker and Chemla 2018 show that some gestural verbs display agreement-like properties that are strikingly reminiscent of ASL agreement verbs, including under ellipsis.

To address this alternative interpretation of the data, we turn to cases in which the target classifier is more deeply embedded and couldn't involve agreement.

### 3.4 Action Role Shift triggers viewpoint shift: complex constructions

In the following constructions (structurally analogous to the gestural example in (16)), Role Shift applies over a multi-predicate clause, and the classifier pertains not to the character that triggers Role Shift, but to an animal that interacts with that character. Obviously the animal classifier doesn't have the character as a subject and thus agreement plays no role in these examples. But as we will see, Role Shift still triggers viewpoint shift (to the character's viewpoint) in the interpretation of the animal classifier.

In (25) (already discussed in simplified form in (1)), the signer establishes a locus *b* for Ann and her circus and a locus *a* for his own zoo, and he asserts that Ann ran away from her initial position, in a straight path displayed by a person classifier, as shown in (26)a. The target sentence is a multiclausal construction with a classifier, without Role Shift in (25)a, and with Action Role Shift in (25)b (the fact that Action rather than Attitude Role Shift is involved is clear from the meaning: no words or thoughts are reported).

- (25) YESTERDAY ANN PERSON-cl<sub>b</sub>, EYES ITCH. AREA<sub>b</sub> POSS-b CIRCUS. AREA<sub>a</sub> POSS-1 ZOO  
'Yesterday, Ann's eyes were itching. She was at her circus. Further away was, my zoo,  
NONE FENCE. IX-b b-RUN-a. IX-b b-PERSON-a-cl.  
which has no fence. Ann was running towards my zoo.

scared eyes\_\_\_\_\_



- a. <sup>6.3</sup>JOG PALMS-UP EYES-DROP, SCARED, ELEPHANT [a-ANIMAL-b-cl] \_\_\_\_\_  
 [Ambiguous] She stopped running to put drops in her eyes and she got scared by an elephant **coming (i) from the front (on the right) [3/3 judgments] or (ii) from the rear right side [2/3 judgments].**

RS<sub>b</sub>\_\_\_\_\_

scared eyes\_\_\_\_\_

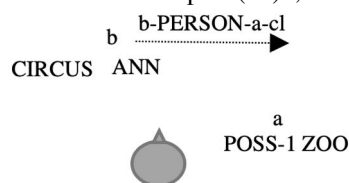


- b. <sup>6.7</sup>JOG PALMS-UP EYES-DROP, SCARED, ELEPHANT [a-ANIMAL-cl-b] \_\_\_\_\_  
 [Unambiguous] She stopped running to put drops in her eyes and she got scared by an elephant **coming from the right [3/3 judgments].**

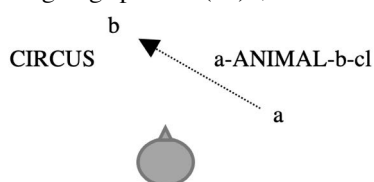
(ASL 37, 3480, 3 judgments; <https://youtu.be/DXOoLe2j3wI>)

(26) Loci and classifier movement in (25)

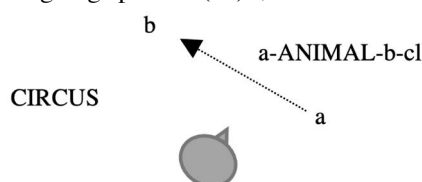
- a. Context of examples (25)a,b



- b. Signing space in (25)a, no Role Shift



- b. Signing space in (25)b, Role Shift



In the absence of Role Shift, there is an ambiguity. The elephant classifier movement, displayed in (26)b, can be interpreted relative to the signer's initial viewpoint, and thus relative to the same viewpoint as the classifier representing Ann's movement in (26)a. On this absolute interpretation, the elephant approaches Ann from the front (right side). Alternatively, the elephant movement can be interpreted relative to a shifted viewpoint. On this interpretation, the "camera position" is associated with Ann, and the elephant movement relative to the signer in (26)b corresponds to what Ann perceives, hence the elephant is coming from the rear (right side). Strikingly, with Role Shift as in (25)b, only the interpretation with viewpoint shift is available. Given the signer's role-shifted position, the elephant comes from the right, more or less orthogonally cutting Ann's path. Here in two separate sessions, our second consultant initially interpreted both sentences without viewpoint shift, and then took the non-role-shifted sentence to be ambiguous and the role-shifted sentence to involve viewpoint shift (see Supplementary Materials C).

We turn to a paradigm in which Ann's orientation relative to her surrounding is established thanks to a sign representing rows on which Ann's students are sitting. In (27), the target classifier represents a cat, whose movement may be interpreted with or without viewpoint shift (the loci and classifiers are represented in (28), with a human figure added to represent Ann, in order to help the reader see how this configuration is interpreted).

- (27) YESTERDAY POSS-1 CAT IN CLASS. ANN<sub>b</sub> TEACHER EYES ITCH. STUDENT b-ROW-a-cl.  
 ANN<sub>b</sub> IX-b

'Yesterday, my cat was in class. Ann was the teacher, and her eyes itched. The students were in rows in front of her. As Ann

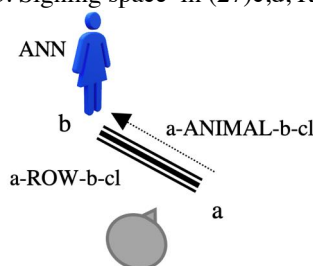
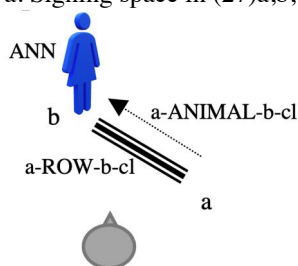


- a. <sup>5.7</sup> TEACH PALMS-UP EYES-DROP SAME-TIME CAT a-ANIMAL-b-cl\_ taught, she put in eye drops. At the same time, the cat moved **(i) diagonally towards Ann, back to front, right to left [3/3 judgments], or (ii) towards Ann [3/3 judgments]**.
- b. <sup>6</sup> TEACH PALMS-UP EYES-DROP SAME-TIME :o) CAT a-ANIMAL-b-cl. taught, she put in eye drops. At the same time, she was surprised to see that the cat moved **(i) diagonally towards Ann, back to front, right to left [3/3 judgments], or maybe (ii) towards Ann [1/3 judgment]**
- RS \_\_\_\_\_



- c. <sup>6</sup> TEACH PALMS-UP EYES-DROP SAME-TIME CAT a-ANIMAL-b-cl\_ (7, 7, 4) taught, she put in eye drops. At the same time, the cat moved **in front of her, right to left.**
- RS \_\_\_\_\_
- d. <sup>6.7</sup> TEACH PALMS-UP EYES-DROP SAME-TIME :o) CAT a-ANIMAL-b-cl. taught, she put in eye drops. At the same time, she was surprised to see that the cat moved **in front of her, right to left.**
- (ASL 37, 3206, 3 judgments; <https://youtu.be/BSmwAdyTAgY>)

- (28) a. Signing space in (27)a,b, no Role Shift      b. Signing space in (27)c,d, Role Shift



The results are in line with the Viewpoint Shift Generalization in (3)a. Without Role Shift, in (27)a,b, the cat movement may but need not be interpreted relative to Ann's viewpoint. The addition of an expression of surprise in (27)b makes dominant the interpretation corresponding to what Ann saw (hence with viewpoint shift). With Role Shift, as in (27)c,d, the cat classifier movement relative to the signer has to represent the cat's movement relative to Ann's viewpoint: the construction must be interpreted with viewpoint shift. Here our second consultant interpreted all examples with viewpoint shift, which is consistent with (but does not prove) the Viewpoint Shift Generalization, as the latter allows for ambiguity of non-role-shifted sentences.

### 3.5 Intermediate summary and comparison with our second consultant's judgments

In sum, we have seen by way of several constructions that, for our main consultant, Action Role Shift implies viewpoint shift. First, bare classifiers under Role Shift must start from the signer's role-shifted position; this applies when viewpoint shift involves both the horizontal and the vertical dimension. Second, when Role Shift targets a clause with an embedded classifier, its movement must be depicted from the viewpoint associated with the character that triggers Role Shift.

Our second consultant's judgments are more complex. For him, Role Shift implied viewpoint shift in the horizontal dimension, but not always in the vertical dimension, as shown by his acceptance of (24)a. However he found the expected contrast with height in an example discussed in Appendix I-A (where our main consultant found no contrast, possibly because he applied Role Shift twice). For complex constructions, our second consultant partly displayed the expected contrast in (25) (ambiguity without Role Shift, viewpoint shift with Role Shift, but only in his second session<sup>15</sup>). For (27), he

<sup>15</sup> As mentioned, he obtained no viewpoint shift in either example in his first session.

obtained viewpoint shift both with and without Role Shift, which is consistent with our analysis, but of course does not prove it. The partial disagreement between the two consultants clearly calls for more empirical work, a point we come back to in the conclusion.

## 4 Attitude Role Shift and Viewpoint Shift

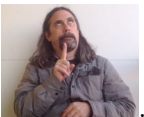
We will now extend our investigation to classifiers that are embedded under Attitude Role Shift (still focusing on the detailed and repeated judgments of our main consultant). As stated in generalization (3)b, the entire role-shifted clause must depict a signing event viewed from the shifted viewpoint. This is just the old observation that Attitude Role Shift comes with a strong quotational component (e.g. Lillo-Martin 1995, Quer 2011, Schlenker 2017b and Davidson 2015). On the other hand, individual classifiers may but need not be evaluated with respect to the attitude holder's shifted viewpoint. Since we do not reiterate the argument for the strongly quotational nature of Attitude Role Shift, our paradigms below display the *absence* of a generalization: classifiers may but need not be interpreted relative to the attitude holder's viewpoint. And since we just aim to highlight the interpretive freedom of classifiers, we discuss the main data more briefly than for Action Role Shift, providing some complements in Appendix I-B. (As announced, we do not discuss gestural data in this section. Still, it is worth noting that the self-pointing gesture that appeared under Gestural Attitude Role Shift in (11)(ii) above is rather clearly quoted, although it is a single expression. In Appendix III, we discuss more complex gestural constructions that appear to be quoted under Gestural Attitude Role Shift.)

### 4.1 Attitude Role Shift and viewpoint shift: classifiers

We explored the same paradigm types as in Section 3: immediate embedding of classifiers moving in the horizontal and in the vertical dimensions, and deeper embedding of additional classifiers.

Starting with immediate embedding involving the horizontal dimension, (29) is the counterpart with an attitude report of an earlier example pertaining to an action report, namely (20). The presence of a full first person pronoun under Role Shift, as is the case in (29)d-f, shows that Attitude rather than Action Role Shift is involved. Role Shift triggers viewpoint shift, as seen by the relative deviance of (29)d (although judgments were unstable). Importantly, the shifted viewpoint is *not* the viewpoint associated with the context of the reported speech act, as Ann was probably not making her statement while walking away from school.<sup>16</sup> On the other hand, something close to Ann's very signs is iconically reproduced, since Attitude Role Shift has a strong quotational component.

- (29) TOMORROW ANN<sub>a</sub> THERE-b COLLEGE. POSS-a HOME NEAR IX-c. IX-b WILL FIRE WARN  
 'Tomorrow Ann will be at the college. Her home is not far. There will be a fire alarm  
 PRACTICE. IX-1 NOT THERE. IX-a SAY  
 drill. I won't be there, but she says

|                           |                                                 |                                                                                     |
|---------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------|
|                           | leisurely_____                                  |  |
| a. <sup>7</sup> IX-a WILL | b-PERSON-c-cl_                                  | .                                                                                   |
|                           | leisurely_____                                  |  |
| b. <sup>7</sup> IX-a WILL | 1-PERSON-c-cl_                                  | .                                                                                   |
| c. <sup>7</sup> IX-a WILL | 1-PERSON-c-cl_ <sub>forward_body_movement</sub> | .                                                                                   |

<sup>16</sup> Technically, in standard De Se analyses of attitude reports, such as the one we develop in Section 6, what matters is not the context of the agent's attitude, but rather the contexts compatible with that attitude. This does not change the picture: the contexts compatible with Ann's claim are ones in which, *in the past*, she hesitated and jumped. It is not the viewpoint of these contexts which is adopted here, but rather the past viewpoint associated with Ann.

she will leave leisurely and go home.'

- RS<sub>a</sub>\_\_\_\_\_
- leisurely\_\_\_\_\_
- d. <sup>4,7</sup> IX-1 WILL b-PERSON-c-cl\_  (7, 5, 2)
- RS<sub>a</sub>\_\_\_\_\_
- leisurely\_\_\_\_\_
- e. <sup>6,7</sup> IX-1 WILL 1-PERSON-c-cl\_ 
- RS<sub>a</sub>\_\_\_\_\_
- f. <sup>7</sup> IX-1 WILL 1-PERSON-c-cl<sub>forward\_body\_movement</sub>.

: I will leisurely leave and go home.'

(ASL 37,3354, 3 judgments; <https://youtu.be/kxEzIEi7jds>)

(30) Loci and classifier movement in (29)

- a. Example (29)a
- HOME<sub>c</sub> ← PERSON-cl b COLLEGE HOME<sub>c</sub>
- a ANN
- b. Examples (29)b,c
- HOME<sub>c</sub> ↑ PERSON-cl b COLLEGE
- a ANN
- c. Example (29)d
- HOME<sub>c</sub> ← PERSON-cl b COLLEGE
- a ANN
- d. Examples (29)e,f
- HOME<sub>c</sub> ↓ PERSON-cl b COLLEGE
- a ANN

Since the viewpoint of the attitude holder matters for the quotational interpretation of the role-shifted clause but not for the interpretation of classifiers that appear in it, we might expect that classifiers have the interpretation they would have in direct discourse. In other words, we might expect that the same viewpoints of evaluation are used to interpret classifiers under Attitude Role Shift and in direct discourse. This expectation is not met, as the direct discourses in (31) are all acceptable (see Appendix I-B1 for a further examples that involve the horizontal dimension).

(31) TOMORROW IX-1 THERE-a COLLEGE. POSS-1 HOME NEAR IX-b. IX-a WILL FIRE WARN PRACTICE.

Tomorrow I will be at the college. My home is nearby. There will be a fire alarm drill.

- leisurely\_\_\_\_\_
- a. <sup>7</sup> IX-1 WILL a-PERSON-b-cl\_ 



leisurely\_\_\_\_\_

b. <sup>6.7</sup> IX-1 WILL 1-PERSON-b-cl\_\_\_\_\_.

leisurely\_\_\_\_\_

c. <sup>7</sup> IX-1 WILL 1-PERSON-b-cl<sub>forward\_body\_movement</sub>.  
I will leisurely leave and go home.(ASL 37, 3358, 3 judgments; [https://youtu.be/vtVCLfFZ\\_3s](https://youtu.be/vtVCLfFZ_3s))

Turning to the vertical dimension, in (32), the action report discussed in (24) is replaced with an attitude report. Role Shift again implies viewpoint shift, but here too the viewpoint is not that of the attitude holder Ann, as she is certainly not recounting her story from the side of a springboard. Direct discourse only displays a weak preference for the same shifted viewpoint, as seen in (32). (Here our second consultant mentioned that he would score both examples in (32) as 7. An additional example with the same structure can be found in Appendix I-B1.)

(32) YESTERDAY ANN INVOLVE DIVE COMPETITION. HAVE HIGH PLANK, PLANK-cl<sub>a high</sub>.  
'Yesterday, Ann took part in a diving competition. There was a spring board high up.

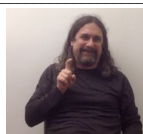
RECENTLY ANN SAY

Recently, Ann said

enthusiastic\_\_\_\_\_

a. <sup>7</sup> IX-a a<sup>high</sup>-PERSON-stands-then-jumps-low-cl\_\_\_\_\_.

enthusiastic\_\_\_\_\_

b. <sup>6.3</sup> IX-a 1-PERSON-stands-then-jumps-low-cl\_\_\_\_\_.  
that she enthusiastically stood there and jumped.'RS<sub>a</sub>\_\_\_\_\_

enthusiastic\_\_\_\_\_

c. <sup>3.3</sup> IX-1 1-PERSON-stands-then-jumps-low-cl\_\_\_\_\_.RS<sub>a</sub>\_\_\_\_\_

enthusiastic\_\_\_\_\_

d. <sup>7</sup> IX-1 a<sup>mid</sup>-PERSON-stands-then-jumps-low-cl\_\_\_\_\_.  
: I enthusiastically stood there and jumped.'(ASL 37, 3458, 3 judgments; <https://youtu.be/l0jE5I-xpj4>)



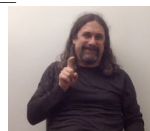
- (33) *Context*: Yesterday the signer took part in a high diving competition. The plank was high.

enthusiastic\_\_\_\_\_



- a. <sup>6</sup> IX-1 a<sup>high</sup>-PERSON-stands-then-jumps-low-cl\_ .

enthusiastic\_\_\_\_\_



- b. <sup>7</sup> IX-1 1-PERSON-stands-then-jumps-low-cl\_ .

'I enthusiastically stood there and jumped.'

(ASL 37, 3472, 3 judgments; <https://youtu.be/-HUVjrK1jb4>)

Turning to complex embedded constructions, the example in (34) (with the classifier movement in (35)) shows that, with Role Shift, the viewpoint must be shifted. The interpretation of the first person possessive *POSS-1* is a clear correlate of Role Shift or of its absence, as it denotes the signer in (34)a and the attitude-holder Ann in (34)b. Importantly, there could in principle be an ambiguity in (34)a, which does not involve Role Shift. The reason is that part of the target clause following *SAY* could be unembedded, and asserted by the signer. The final words 'I don't know whether what she said is true' are intended to force embedding by making it clear that what follows *SAY* is attributed to Ann. (Here our second consultant had a similar acceptability contrast, and understood both sentences with viewpoint shift.)

- (34) *Context*: Ann often tells surprising stories. They may or may not be true, depending on the case.

YESTERDAY ANN PERSON-cl<sub>b</sub>, EYES ITCH. AREA<sub>b</sub> POSS-b CIRCUS. AREA<sub>a</sub> POSS-1 ZOO

'Yesterday, Ann's eyes were itching. She was at her circus. Further away was my zoo,

NONE FENCE. IX-b b-RUN-a. IX-b b-PERSON-a-cl. IX-b

which has no fence. Ann was running towards my zoo. She

a. <sup>5.3</sup> SAY JOG PALMS-UP EYES-DROP, SCARED, POSS-1 ELEPHANT [a-ANIMAL-b-cl].

[Ambiguous] said that she stopped running to put drops in her eyes and she got scared by my elephant **coming from the front (on the right) / crossing her path orthogonally from the right.**

RS<sub>b</sub>\_\_\_\_\_

b. <sup>6.3</sup> SAY JOG PALMS-UP EYES-DROP, SCARED, POSS-1 ELEPHANT [a-ANIMAL-cl-b].

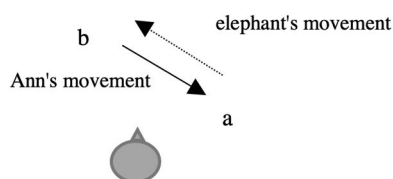
[Unambiguous] said: I stopped running to put drops in my eyes and I got scared by my elephant **coming from the front (on the right).**

POSS-b STORY TRUE NOT-KNOW.

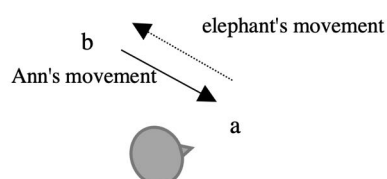
I don't know whether what she said is true.'

(ASL 37, 3492, 3 judgments; <https://youtu.be/GErGUCrLKYw>)

- (35) a. Movements without Role Shift



- b. Movements with Role Shift



As can be seen, with Role Shift the viewpoint of interpretation is associated with the agent of the reported action. Importantly, this need not be the viewpoint corresponding to Ann as she tells the story. (An additional example with the same structure can be found in Appendix I-B2.)

Summarizing, in the examples we have seen so far, Attitude Role Shift comes with a shifted viewpoint, but this viewpoint is not that of that attitude holder, i.e. Ann telling her story after the fact. Rather, the viewpoint is associated with Ann as she experienced the action.

We will now see that there are very clear cases in which the viewpoint of interpretation of role-shifted classifiers is neither associated with the attitude holder nor with the agent of the reported action. Specifically, when direct discourse makes salient entirely different viewpoints, Attitude Role Shift makes them available as well.

#### 4.2 Attitude Role Shift and viewpoint shift: when direct discourse makes available additional viewpoints

To introduce Iconological Semantics in Section 2.1, we discussed some cases in which viewpoint variables do not denote the signer's viewpoint. When these examples are embedded under Attitude Role Shift, they yield clear cases in which the viewpoint of evaluation of embedded classifiers is different from the viewpoint of the attitude holder, and also of the agent of the reported action.

In (36), the sentence in (7), with quantification over viewpoints, is embedded under SAY, and it is clear that the viewpoint of evaluation of the embedded classifiers is not that of the attitude holder (the lead teacher), but rather a quantified viewpoint as in (7).

- (36) *Context:* The lead teacher's student group has classes in 4 classrooms, one for each of 4 teachers under the lead teacher's authority (each teacher always teaches in the same classroom).  
 LEAD TEACHER SAY  
 'The lead teacher says  
 a. <sup>7</sup>POSS-a GROUP CLASS BREAK ALL ALWAYS HAVE STUDENT PERSON-walk-back\_right-cl.  
 in his group, in each class break, there's always a student that leaves toward the back, to the right.'  
 b. <sup>7</sup>POSS-a GROUP CLASS BREAK ALL ALWAYS HAVE STUDENT PERSON-walk-right->left-cl.  
 in his group, in each class break, there's always a student that leaves toward the left, from the right.'  
 RS  
 c. <sup>7</sup>POSS-1 GROUP CLASS BREAK ALL ALWAYS HAVE STUDENT PERSON-walk-back\_right-cl.  
 : In my group, in each class break, there's always a student that leaves toward the back, to the right.'  
 RS  
 d. <sup>7</sup>POSS-1 GROUP CLASS BREAK ALL ALWAYS HAVE STUDENT PERSON-walk-right->left-cl.  
 : In my group, in each class break, there's always a student that leaves toward the left, from the right.'  
 (ASL 37, 3420, 3 judgments; <https://youtu.be/rKP0shku-OA>)

It stands to reason that these examples, both with and without Role Shift, should be analyzed with an embedded version of the Logical Form discussed in (7)b—along the lines of (37) (which is simplified). It is clear that the embedded viewpoint of evaluation is not that of the lead teacher.

- (37) teacher say always<sub>ST</sub>  $\exists \pi$  there-is student person-walk-cl <sup>$\pi$</sup>

Further examples can be constructed in which the viewpoint variable is free, but does not refer to the attitude holder's viewpoint, nor to that of the agent of the reported action. To see how this works, we start from the example without embedding in (38): the classifier is evaluated relative to the signer's viewpoint as he was in class yesterday, not to his current viewpoint.

- (38) *Context:* The conversation is taking place in the library.  
 a. <sup>6,7</sup>POSS-1 INTRO CLASS YESTERDAY BREAK HAVE PRESIDENT PERSON-walk-back\_right-cl.  
 'In my introductory class yesterday, during the break, the president walked towards the back, to the right.'  
 b. <sup>6,7</sup>POSS-1 INTRO CLASS YESTERDAY BREAK HAVE PRESIDENT PERSON-walk-right->left-cl.

cl.

'In my introductory class yesterday, during the break, the president walked towards the left, from the right.'

(ASL 37, 3374, 3 judgments; <https://youtu.be/0ZbOdQICnbM>)

A heavily simplified Logical Form appears in (39), where the viewpoint variable  $\pi$  does not refer to the signer's current viewpoint, but to his viewpoint yesterday.

(39) there-is president person-walk-cl <sup>$\pi$</sup>

Unsurprisingly, when this sentence is embedded under *SAY*, the salient viewpoint remains the same, and is distinct from the viewpoint of the attitude holder at the time of the saying (the boldfaced part of the context is intended to make it unlikely that what follows *WHAT* is attributed to the signer; rather, the entire following material must be attributed to the teacher, not to the signer). This viewpoint is also distinct from that of the agent of the reported action, namely the university president.

(40) *Context:* The conversation is taking place outside of school  
RECENTLY LINGUISTICS TEACHER<sub>a</sub> THE-TWO<sub>1,a</sub> IN LIBRARY. **IX-1 NOT BELIEVE IX-a.**  
'Recently the linguistics teacher and I were in the library. I don't believe him, specifically this.

WHAT? IX-a SAY

a. <sup>7</sup> POSS-a INTRO CLASS YESTERDAY, BREAK, HAVE PRESIDENT PERSON-walk-back->right-cl.

He said that in his introductory class yesterday, during the break, the president entered from the back door of the class on the right, and exited on the same side, in the front.'

b. <sup>7</sup> POSS-a INTRO CLASS YESTERDAY BREAK HAVE PRESIDENT PERSON-walk-right->left-cl.

He said that in his introductory class yesterday, during the break, the president crossed the room from right to left.'

RS

c. <sup>7</sup> POSS-1 INTRO CLASS YESTERDAY BREAK HAVE PRESIDENT PERSON-walk-back->right-cl.

He said: 'In my introductory class yesterday, during the break, the president entered from the back door of the class on the right, and exited on the same side, in the front.'

RS

d. <sup>7</sup> POSS-1 INTRO CLASS YESTERDAY BREAK HAVE PRESIDENT PERSON-walk-right->left-cl.

He said: 'In my introductory class yesterday, during the break, the president crossed the room from right to left.'

(ASL 37, 3370, 3 judgments; <https://youtu.be/w1kPx7xbRG0>)

### 4.3 *Intermediate summary and comparison with our second consultant's judgments*

We might have expected that in Attitude Role Shift, the viewpoint of evaluation is that of the attitude holder. This is not so at all. The examples of Section 4.1 have a shifted viewpoint associated with the agent of the reported action, not with the attitude holder. But this is not the only possibility. In the examples of Section 4.2, direct discourse involves quantified or salient viewpoints, and this property is preserved under Attitude Role Shift, including when the shifted viewpoint is not associated with the agent of the reported action (we do not know why the connection with direct discourse is far less strict in the examples of Section 4.1).

The general picture is thus one of great freedom, which contrasts with what we found for Action Role Shift. But this does not mean that the viewpoint of the attitude holder doesn't play a crucial role. Attitude Role Shift has been shown to come with a strong quotational component, and thus the signs used in the embedded clause are a representation of the attitude holder's own signs, evaluated from the attitude holder's perspective. This will yield the key to the apparent contrast between Action Role Shift and Attitude Role Shift.

It is worth noting that, as shown in the Supplementary Materials C, our second consultant's contrasts generally matched those of our main consultant (the main exception is that he understood both examples in (34) with viewpoint shift).

We will now develop two analyses of Role Shift that crucially involve viewpoint shift. In the simplest theory, Role Shift is just visible viewpoint shift—what the signer's shifting signals is that he is adopting another viewpoint, and all the words that are signed in shifted position must be evaluated relative to the shifted viewpoint. This will account for the fact that the viewpoint of evaluation is shifted under Action Role Shift, and that the signs are quoted in Attitude Role Shift. But our analysis will not account for the fact that indexicals are usually blocked under Action Role Shift, and that wh-extraction is possible under Attitude Role Shift. Deriving these results will require a richer theory in which Role Shift involves context+viewpoint shift.

## 5 Role Shift as Overt Viewpoint Shift

### 5.1 Main ideas

The simplest theory of Role Shift treats it as an overt instantiation of viewpoint shift. On this analysis, the signer signals that he is adopting another viewpoint, for instance by rotating his body, shifting his eyegaze, and adopting the facial expressions of someone associated with that other viewpoint. As long as the signer signals that he is in a shifted position, his signs are evaluated relative to the shifted viewpoint. To make these ideas precise, we posit the principle in (41):

(41) **Role Shift as overt viewpoint shift**

If the signer role-shifts to adopt a certain viewpoint, every iconic expression signed under Role Shift must be indexed with a variable that denotes that viewpoint.

This condition will immediately account for Action Role Shift. For Attitude Role Shift, it will have to be combined with the assumption that the role-shifted clause is quoted (e.g. Davidson 2015), and thus behaves like a single expression, opaque to grammatical operations. On this analysis, the embedded clause as a whole is the iconic representation of a signing event.

Why should Role Shift trigger viewpoint shift? A natural idea, in line with informal and formal views of sign language iconicity (e.g. Liddell 2003, Perniss 2007, Schlenker and Lamberton 2024), is that the signer as puppeteer is faced with the addressee as an audience. The signer's body shift and eyegaze shift help indicate that another viewpoint is adopted to recreate the scene. The addressee must thus mentally put themselves in a position that faces the shifted body and eyegaze, and adopt the corresponding viewpoint as well (from the audience side of the puppet show, so to speak).<sup>17</sup>

### 5.2 Action Role Shift

Let us see how (41)a derives the main properties of Action Role Shift. Consider first one of our simplest examples, (10)a, copied in (42)a. It can be given the Logical Form in (42)b. Since Role Shift is to position *a*, the role-shifted classifier must be indexed with a viewpoint variable  $\pi_a$  that denotes the corresponding viewpoint (note that we use  $\pi_a$  rather than  $\pi_k$  for mnemonic purposes, but the conditions only has to do with the denotation of the viewpoint variable, not with its form<sup>18</sup>). This condition is satisfied in (42)b. As is standard, here and throughout, we write *word'* (e.g. *person'*) for the semantic value of an expression *word* (e.g. *person*).

*Notation:* To avoid having subscripts of superscripts, we write  $\pi_a$  as  $\pi_{-a}$  when it appears as a superscript.

- (42)                      RS<sub>a</sub>\_\_\_\_\_
- a. IX-a                1-WALK-WITH-ENERGY(CL-ONE).  
'He left with energy.' (ASL, 14, 233)
- b. pro<sub>a</sub> person-cl <sup>$\pi_{-a}$</sup>
- c. (b) satisfies (41)a if  $s(\pi_a)$  is a viewpoint associated with  $s(a)$ . If so,  
[[[b]]]<sup>c, s, t, w</sup> = [[person-cl] <sup>$\pi_{-a}$</sup> ]<sup>c, s, t, w</sup> ([[pro<sub>a</sub>]]<sup>c, s, t, w</sup>)

<sup>17</sup> In the refinement discussed in Section 5.4, the shifted viewpoint is associated with an individual. The signer's facial expressions may help depict the individual triggering Role Shift.

<sup>18</sup> Concretely, nothing would change if we had picked a variable  $\pi_k$ —the condition is that  $s(\pi_k)$  should be associated with  $s(a)$ .

$$\begin{aligned}
&= [\lambda x_e \cdot \text{person}'_{t,w}(x) = 1 \text{ and } \text{proj}(x, s(\pi_a), t, w) = \text{person-cl}](s(a)) \\
&= 1 \text{ iff } \text{person}'_{t,w}(s(a)) = 1 \text{ and } \text{proj}(s(a), s(\pi_a), t, w) = \text{person-cl}
\end{aligned}$$

If  $s(\pi_a)$  is the viewpoint associated with the initial position of individual  $s(a)$  as it started moving, we obtain appropriate truth conditions: the person's movement relative to that viewpoint should be as the classifier's movement relative to the signer's shifted position. It is worth noting that, from the present perspective, the fact that the classifier starts out from the signer's role-shifted position need not be a reflex of a first person agreement marker. Rather, this is what is expected on iconic grounds because the person starts moving from the position of the shifted viewpoint.

This analysis makes it possible to analyze the paradigm in (18) above. There are two Logical Forms to consider. In (43)a, the classifier is evaluated with respect to a viewpoint that is unrelated to that of the agent, whereas in (43)b the viewpoint of evaluation corresponds to the agent's initial position. In (18)a,b, there is no overt viewpoint shift and thus both Logical Forms can be employed. In (18)c,d, by contrast, Role Shift instantiates overt viewpoint shift, and correspondingly the viewpoint of evaluation must be associated with the person that triggers Role Shift. As a result, only (43)b is acceptable.

- (43) a.  $\text{pro}_a \text{ person-cl}^{\pi_i}$ , with  $s(\pi_i)$  unrelated to  $s(a)$   
 b.  $\text{pro}_a \text{ person-cl}^{\pi_a}$ , with  $s(\pi_a)$  corresponding to  $s(a)$ 's initial position

The same analysis applies to cases involving the vertical dimension, as in the paradigm in (24) above. If  $\pi_i$  denotes the original viewpoint and  $\pi_a$  denotes a viewpoint associated with Ann's initial position, Role Shift to Ann's position makes it mandatory to have the viewpoint choice in (44)b. This explains the deviance of the role-shifted clause in (24)c above, as it involves the Logical Form in (44)a.

- (44) a.  $\text{Ann person-stands-then-jumps-cl}^{\pi_i}$ , with  $s(\pi_i)$  unrelated to  $s(a)$   
 b.  $\text{Ann person-stands-then-jumps-cl}^{\pi_a}$ , with  $s(\pi_a)$  corresponding to  $s(a)$ 's initial position

There is nothing in the analysis to prohibit the occurrence of several expressions under Action Role Shift, although the constraint in (41)a implies that each iconic expression should be indexed with a viewpoint associated with the shifted position. In this way, the sentence in (25)b above can be analyzed as in (45)b but not as in (45)a. The problem with the latter Logical Form lies with the boxed part: the viewpoint corresponding to the animal classifier is not associated with Ann (denoted by the pronoun  $\text{pro}_b$ ). (We posit a null pronoun  $\emptyset_b$  denoting Ann ( $= s(b)$ ), and we assume, as is standard for ASL, that conjunctions can be covert.)

- (45) a.  $\emptyset_b \text{ palms-up} \ \& \ \emptyset_b \text{ eyes-drop} \ \& \ \text{scared} \ \& \ \text{elephant animal-move-cl}^{\boxed{\pi_i}}$ , where  $s(\pi_i)$  is a viewpoint that is not associated with  $s(b)$   
 b.  $\emptyset_b \text{ palms-up} \ \& \ \emptyset_b \text{ eyes-drop} \ \& \ \text{scared} \ \& \ \text{elephant animal-move-cl}^{\pi_b}$ , where  $s(\pi_b)$  is a viewpoint associated with  $s(b)$

We derive in (46) the meaning of the last conjunct of (45)b. It can then be combined with the meanings of the preceding conjuncts.

- (46)  $[[\text{elephant animal-move-cl}^{\pi_b}]]^{c,s,t,w} = [[\text{animal-move-cl}^{\pi_b}]]^{c,s,t,w} ([[\text{elephant}]]^{c,s,t,w})$   
 $= [\lambda x_e \cdot \text{animal}'_{t,w}(x) = 1 \text{ and } \text{proj}(x, s(\pi_b), t, w) = \text{animal-move-cl}](\text{elephant}')$   
 $= 1 \text{ iff } \text{animal}'_{t,w}(\text{elephant}') = 1 \text{ and } \text{proj}(\text{elephant}', s(\pi_b), t, w) = \text{animal-move-cl}$

In sum, analyzing Action Role Shift as overt viewpoint shift has the advantage of explaining the main target data. A further question, which we leave for future research, is what justifies Action Role Shift—presumably something iconic, as emphasized by the literature (see for instance Davidson 2015, Schlenker 2017a,b and Waller 2021). Finding the right formulation of this condition is an issue we leave for future research.<sup>19</sup>

<sup>19</sup> Two potential formulations appear in (i). (i)b seems too strong (e.g. Schlenker 2017a has examples with *FRIEND*, with no iconic modification, under Action Role Shift). (i)a might be too weak.

- (i) Additional conditions on Action Role Shift as overt viewpoint shift  
 a. At least one expression found under Action Role Shift should be iconic.  
 b. Every expression found under Action Role Shift should be iconic.

### 5.3 Attitude Role Shift

Let us turn to an analysis of Attitude Role Shift as overt viewpoint shift. On the present theory, the embedded clause is most naturally analyzed as a quotation. We will explain how a quotational reading can fulfill the requirements of viewpoint shift. On the other hand, we will leave a question open, namely how an *indirect* discourse reading with viewpoint shift could be blocked on principled grounds.

The main task is to analyze quotation within Iconological Semantics, as it is clear that a signed quotation is a three-dimensional recreation of a signing event. Following insights in Davidson 2015, we propose to analyze the meaning of a quotation by analogy with classifier predicates: a quotation evaluated from a certain viewpoint is true of all signing events that project to this quotation from the relevant viewpoint. The rule is stated in (47), where  $v$  is the type of events.

$$(47) \llbracket \text{quotation}^{\pi} \rrbracket^{c, s, t, w} = \lambda e_v . \text{signing-event}'_{t, w}(e) = 1 \text{ and } \text{proj}(e, s(\pi), t, w) = \text{quotation}$$

Consider for instance the end of the sentences in (29)e, repeated in (48).

$$(48) \quad \text{RS}_a \text{_____} \\ \text{IX-a SAY IX-1 WILL } \text{leisurely} \text{_____} \\ \text{1-PERSON-c-cl.}$$

On the present analysis, the embedded clause is entirely quoted and functions (as quotations do) as a single expression, something that we write with  $\_$  between words, as in (49)a. Furthermore, only overt material is quoted, and correspondingly we do not include covert material (such as viewpoint variables) in the quoted sentence.

$$(49) \text{ a. } \text{pro}_a \text{ say } [\text{pro}_1 \text{ will person-cl}^{\wedge} \text{leisurely}]^{\pi_a} \\ \text{b. } [\exists [\text{pro}_1 \text{ will person-cl}^{\wedge} \text{leisurely}]^{\pi_a}] \lambda e_v \text{ pro}_a \text{ say } e$$

Since a sign quotation is true of multiple signing events it can depict, we assume that the Logical Form of the sentence contains an existential quantifier, as depicted in (49)b. As is standard (e.g. Heim and Kratzer 1998), an existential quantifier (with its restrictor) moves at LF for type reasons, giving rise to the representation in (49)b. The semantics of the existential quantifiers, which we take to be cross-categorical, is given in (50); it is standard (here  $\bullet$  stands for any type).

$$(50) \llbracket \exists \rrbracket^{c, s, t, w} = \lambda f_{\langle \bullet, \triangleright \rangle} \lambda g_{\langle \bullet, \triangleright \rangle} . \text{for some } x \text{ in } D, f(x) = g(x) = 1$$

Schematically, then, the target Logical Form to be interpreted is in (51)a, where  $D_v$  is the domain of events.

$$(51) \text{ a. } [\exists \text{ quotation}^{\pi_a}] \lambda e_v \text{ pro}_a \text{ say } e \\ \text{b. } \llbracket (a) \rrbracket^{c, s, t, w} = \llbracket \exists \rrbracket^{c, s, t, w} (\llbracket \text{quotation}^{\pi_a} \rrbracket^{c, s, t, w} ([\lambda e_v \text{ pro}_a \text{ say } e])^{c, s, t, w}) \\ = 1 \text{ iff for some } e \text{ in } D_v, \llbracket \text{quotation}^{\pi_a} \rrbracket^{c, s, t, w}(e) = 1 \text{ and } [\lambda e_v \text{ pro}_a \text{ say } e]^{c, s, t, w}(e) = 1 \\ = 1 \text{ iff for some } e \text{ in } D_v, \text{signing-event}'_{t, w}(e) = 1 \text{ and } \text{proj}(e, s(\pi_a), t, w) = \text{quotation and say}'_{t, w}(e)(s(a)) \\ \text{In words: For some signing event } e \text{ that projects to the role-shifted clause from viewpoint } s(\pi_a), s(a) \text{ says } e \\ \text{at } t \text{ in } w$$

When this derivation is applied to the Logical Form in (49)b, two interesting results follow. First, the entire role-shifted clause must be quoted, complete with the facial expressions that accompany it. Second, because projection rules only pertain to *overt* material, there is no constraint on the indices that appear in the role-shifted clause. In fact, the interpretation procedure is blind to these indices, since only overt material can affect the projection function. As a consequence, we do not predict that viewpoint shift should affect the classifiers that appear in the role-shifted clause, which accounts for their great freedom.

On the present theory, it remains to be explained why the role-shifted clause couldn't be analyzed without quotation, namely with standard indirect discourse. This would for instance give rise to the Logical Form in (52) for (48).

$$(52) \text{ pro}_a \text{ say } [\text{pro}_1 \text{ will person-cl}^{\wedge} \text{leisurely}]^{\pi_a}$$

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As pointed out by a reviewer, similar conditions could be explored for Gestural Role Shift.

This Logical Form predicts entirely incorrect results, in particular that the embedded first person pronoun should denote the signer, not the attitude holder (since the present analysis does not countenance context shift). Thus this Logical Form must be blocked. One idea that comes to mind is that a quotational reading is preferred because it is more iconic and thus better justifies the use of viewpoint shift, but we leave the development of such an analysis for future research.

#### 5.4 Refinement: making the individual explicit

Our minimal theory of Role Shift as viewpoint shift can be refined so as to give an explicit role to the person whose viewpoint is adopted. In fact, one could also take  $i$  to be instantiated by a body classifier, as suggested by Davidson 2015, following Supalla 1982. This may be achieved by defining functional viewpoints associated with a person. As defined in (53), if  $i$  is a locus (or more generally a variable, since loci are analyzed as variables),  $\pi_c(i)$  refers to the viewpoint associated with the individual denoted by  $i$  relative to context  $c$ . The relativization to contexts is important because, as we saw at several junctures, Role Shift only specifies that the shifted viewpoint should be closely associated with the individual that triggers the shift, but this still leaves open several choices—it could be the viewpoint of the individual themselves, or one associated with their original position, etc; presumably the context decides. Since individuals can move, we adopt the assumption from, Schlenker et al., to appear, that viewpoints can be dynamic (technically, they may denote different static viewpoints at different times and worlds of evaluation).

- (53) If  $i$  is a locus/variable,  
 $[[\pi(i)]]^{c, s, t, w}$  = the (dynamic) viewpoint associated with  $s(i)$  relative to context  $c$ , which we write as  $\pi_c(s(i))$ .

With this small tweak, the sentence in (42)a can be analyzed as in (54)b.

- (54) **Revised analysis of (42)**  
 a. Original analysis:  $\text{pro}_a \text{ person-cl}^{\pi_a}$   
 b. Revised analysis:  $\text{pro}_a \text{ person-cl}^{\pi(a)}$   
 c. We note that (b) automatically satisfies (41), as by definition if  $[[\pi(a)]]^{c, s, t, w}$  is the viewpoint associated with  $s(a)$ .  

$$[[b)]^{c, s, t, w} = [[\text{person-cl}^{\pi(a)}]]^{c, s, t, w} ([[pro_a]]^{c, s, t, w})$$

$$= [\lambda x_e . \text{person}'_{t, w}(x) = 1 \text{ and } \text{proj}(x, [[\pi(a)]]^{c, s, t, w}, t, w) = \text{person-cl}](s(a))$$

$$= 1 \text{ iff } \text{person}'_{t, w}(s(a)) = 1 \text{ and } \text{proj}(s(a), \pi_c(s(a)), t, w) = \text{person-cl}^{20}$$
*In words:*  $s(a)$  is a person that projects to a person classifier from the viewpoint associated with  $s(a)$  at  $t$  in  $w$

#### 5.5 Limitations

This minimal analysis of Role Shift as overt viewpoint shift has some strengths and some weaknesses.

1. Regarding Action Role Shift, it appropriately predicts that classifiers must be evaluated relative to the shifted viewpoint. But one fact discussed in the literature is not derived. It was claimed on the basis of multiple examples in Schlenker 2017a that indexicals (with the exception of first person agreement markers) are blocked from appearing under Action Role Shift.

Schlenker 2017a explained this observation by a context-shifting analysis of Role Shift. In a nutshell, Action Role Shift affects the context of evaluation of indexicals, and replaces what is initially the context of a communicative or thought event (a "rich context") with an impoverished context that does not correspond to a communicative or thought event. The idea was that indexicals, with the exception of agreement markers, are specified as requiring rich contexts.

From the present perspective, there is no simple way of preventing indexicals from appearing under Action Role Shift. But in fairness, more recent data on the status of first person possessive

<sup>20</sup> Since  $\pi_c(s(a))$  may depend on the time and world of evaluation, we must make use of the revised definition of projection offered in Schlenker et al. 2025, given in (i):

(i) If  $d$  is an object,  $\pi$  a (dynamic) viewpoint,  $t$  a time,  $w$  a world and  $P$  an occurrence of a classifier in signing space,  $\text{proj}(d, \pi, t, w) = P$  iff at  $t$  in  $w$ ,  $d$  projects to  $P$  from  $\pi(t, w)$ , the static viewpoint specified by  $\pi$  at  $t$  in  $w$ .

pronouns under Action Role Shift have *not* confirmed the earlier claim that there is a blanket prohibition against overt indexicals (see Schlenker et al. (to appear), who also speculate that these apparent exceptions might involve interruptions of Role Shift). Further research will have to revisit and possibly refine the earlier generalization.

2. Regarding Attitude Role Shift, our minimal viewpoint shift analysis treats the embedded clause as simply quoted. A positive consequence is that the analysis says nothing about the viewpoint of evaluation of role-shifted classifiers, which indeed seems to display great freedom. On the other hand, quotations are opaque to grammatical dependencies, in essence because the quoted clause is semantically treated as a single object without structure, as suggested by our use of \_ between the quoted words. This raises two potential problems.

(i) First, it was argued in Schlenker 2017a that, despite its quotational properties, ASL Attitude Role Shift allows for wh-extraction (Lee et al. 1997 provide conflicting arguments about ASL). This seems to argue against a pure quotational analysis. But one cautionary note must be added: cases that resemble genuine quotation, involving a sign imitating quotation marks at the beginning of the embedded clause, also appear to allow for wh-extraction (Schlenker 2017a). It could be that despite appearances the embedded clause is not simply quoted in this case; but an alternative is that, for unknown reasons, the consultant allowed for wh-extraction out of quotations quite generally.

(ii) Second, going beyond ASL, it has been argued that in Catalan Sign Language (LSC) and German Sign Language (DGS), Attitude Role Shift is more liberal than in ASL (and LSF) in allowing for the mixing of perspectives in the evaluation of indexicals (Quer 2005, 2013; Herrmann and Steinbach 2012<sup>21</sup>). An example appears in (55): the first person pronoun *IX-I* is evaluated with respect to the reported context (and thus refers to Joan [= *JOAN*]), whereas *HERE* is evaluated with respect to the actual context.

(55)

$$\begin{array}{c} \text{IXa MADRID}_m \text{ MOMENT JOAN}_i \text{ THINK IX-I}_i \text{ STUDY FINISH HERE}_b \\ \text{'When he was in Madrid, Joan thought he would finish his study here (in} \\ \text{Barcelona).'} \end{array}$$

(LSC, Quer 2005, 2013)

If one wishes to extend a viewpoint shift analysis to LSC and DGS, the problem will be to explain why a quoted sentence allows for the mixing of perspectives, something that is not normally permitted. See, however, Maier 2016, 2018 and Hübl et al. 2019 for an analysis of mixed quotation that might help account for such cases.

## 5.6 Comparison with the 'like this' analysis of Role Shift

There are similarities and differences between the viewpoint shift analysis and the theory developed by Davidson (2015), who took the body to function as a classifier in Role Shift. A general similarity is that both analyses take Role Shift to have a strong iconic component, which translates into quotational effects in the case of Attitude Role Shift.

1. The strongest similarity pertains to Attitude Role Shift, which both theories treat as quotational—for better or worse, depending on what one thinks of the wh-extraction data and of the extension to LSC and DGS.

2. With respect to Action Role Shift, there are several important differences.

(i) First, treating the body as a classifier does not by itself derive the Viewpoint Shift Generalization in (3)a. There is a sharp contrast between simple classifiers, which do not trigger viewpoint shift, and Role Shift, which does. On the other hand, the intuition that the signer's body may represent another individual whose perspective is adopted is explicitly taken into account in the refined analysis we developed in Section 5.4.

<sup>21</sup> See also Steinbach 2021, who refers to further examples from Danish Sign Language in Engberg-Pedersen 1995.



(ii) Second, Davidson 2015 claimed that Action Role Shift is limited to elementary predicates, something that is not borne out in our data. She took these predicates to come with a self-referential iconic component. By contrast, the viewpoint shift analysis can apply to a series of words signed under Action Role Shift.

## 6 Role Shift as Overt Context Shift with Viewpoint Shift

We turn to an alternative in which Role Shift simultaneously involves context shift, as argued by Quer 2005, 2013 and Schlenker 2017a,b, and viewpoint shift. This context+viewpoint shift theory derives what was presented as a stipulation in Schlenker 2017a,b. The latter claimed that Role Shift doesn't just shift the context, but also forces the role-shifted expressions to be interpreted maximally iconically. This was one sense in which the Role Shift operator is not a simple "monster", that is, an operator that shifts the context of evaluation of indexicals, but rather a "super monster", which shifts the context and triggers a maximally iconic evaluation. From the present perspective, super monsters should be re-analyzed as operators that simultaneously shift the context and the viewpoint of evaluation.

### 6.1 Main ideas

The intuition we pursue is that by shifting his body and eyegaze and possibly changing his facial expressions, the signer signals that he is mentally putting himself in a different context. This has two consequences: the context parameter is shifted; and the default viewpoint associated with that context is shifted as well. As in the analysis of Role Shift as viewpoint shift, we assume that as long as the signer signals that he is in a shifted position, the viewpoint of evaluation of expressions must correspond to the shifted viewpoint.

Our main assumptions are stated in (56). (56)a implements context shift. (56)b(i) will account for Action Role Shift, while (56)b(ii) will account for Attitude Role Shift: the entire embedded clause will be treated as an iconic expression, and as a result classifiers contained within it will display great freedom in terms of viewpoints of evaluation.

#### (56) Role Shift as overt context shift with viewpoint shift

- If the signer role-shifts to adopt a perspective associated with a context  $c'$ ,
- a. the context of evaluation of expressions signed under Role Shift is  $c'$ ,
  - b. every iconic expression signed under Role Shift must be (i) indexed with the viewpoint of  $c'$ , or (ii) contained within an expression indexed with the viewpoint of  $c'$ .

Why should Role Shift trigger context+viewpoint shift? The explanation is along the same lines as the one we sketched for the pure viewpoint shift theory. The signer plays a dual role as agent of the context and "puppeteer" (for iconic purposes). Shifting his body and eyegaze helps signal that he is mentally putting himself in a different context with a different viewpoint. The addressee will mentally put themselves in the corresponding context and audience position. In addition, for the signer's facial expressions to represent those of another character might help suggest that this other character is the agent of the shifted context.

The implementation goes as follows. We assume (as Schlenker 2017a,b) a framework with a shiftable context parameter of evaluation, in addition to a time, world and assignment function. As is standard, we identify contexts with triples of the form  $\langle \text{agent}, \text{time}, \text{world} \rangle$ . Some contexts (including those introduced by Attitude Role Shift) will correspond to a communicative or thought act, others (introduced by Action Role Shift) will not. If we take role-shifting of a clause to be triggered by a Role Shift operator  $RS$ , its semantics can be given, in the simplest fashion, as in (57): the context of evaluation  $c$  is shifted to a context  $\langle s(a), t, w \rangle$  corresponding to the individual whose perspective is adopted, at the time and world of evaluation (this rule is in the spirit of Anand and Nevins 2004<sup>22</sup>).

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<sup>22</sup> Anand and Nevins 2004 have a simpler intensional system with a context parameter  $c$  and a world parameter  $w$ , which is as finely grained as  $c$  ( $w$  can be taken to be a centered world). Their rule for the context-shifting operator  $Op$  is given in (i):

(57) **Role Shift Operator**

If  $a$  is a locus and  $S$  is a clause,  
 $[[RS_a S]]^{c, s, t, w} = [[S]]^{<s(a), t, w>, s, t, w}$

As in the analysis that treats Role Shift as overt viewpoint shift, we take Role Shift to indicate that the viewpoint of evaluation of role-shifted expressions is the viewpoint associated with the shifted context. To implement this idea, we follow Schlenker et al., to appear, in defining a distinguished ("context-bound") viewpoint variable  $\pi^*$  whose value depends on the context of evaluation. The difference between standard viewpoint variables, of the form  $\pi_i$ , and the distinguished variable  $\pi^*$  can be seen in (58): the value of  $\pi_i$  is given by the assignment function parameter, the value of  $\pi^*$  is given by the context parameter.<sup>23</sup>

(58) **Standard vs. context-bound viewpoint variables** (modified from Schlenker et al., to appear)

- If  $i$  is any index,  
 a.  $[[\pi_i]]^{c, s, t, w} = s(\pi_i)$   
 b.  $[[\pi^*]]^{c, s, t, w} = \pi(c)$ , the (dynamic) viewpoint associated with  $c$

The context-bound viewpoint variable is mandatory under Role Shift, as specified in (59).

(59) **Iconic expressions signed under Role Shift**

Every iconic expression signed under Role Shift must:

- a. carry the context-bound viewpoint variable  $\pi^*$ , or  
 b. be contained within an expression that carries the context-bound viewpoint variable  $\pi^*$ .

As announced, (59)b will account for Attitude Role Shift: the entire role-shifted clause will be treated as a partly iconic expression, and correspondingly classifiers found within it will have great freedom in terms of viewpoint choice. As remarked earlier, one should further develop the theory to explain how much iconicity is sufficient to justify Role Shift (see fn. 19).

It is worth noting that this theory offers greater flexibility than the analysis of Role Shift as simple viewpoint shift. Occasional exceptions to the Viewpoint Shift Generalization in (3) are very hard to explain on the simple viewpoint shift theory. By contrast, the analysis of Role Shift as context+viewpoint shift can in principle allow for cases in which (59) fails to apply, for reasons that would need to be explained. While this greater flexibility is not a good thing for our main consultant, it might be for our second consultant, whose judgments didn't fully adhere to the Viewpoint Shift Generalization.

We will first explain how this analysis accounts for Action Role Shift, and then extend it to Attitude Role Shift, which requires further measures.

## 6.2 Action Role Shift

We start with the same example we analyzed in Section 5, namely (42)a, copied as (60)a. Since we now assume that Role Shift reflects the presence of a context-shifting operator and of a context-bound viewpoint variable  $\pi^*$ , the corresponding Logical Form is as in (60)a'. The boldfaced parts (namely  $RS_a$  and  $\pi^*$ ) distinguish this Logical Form from the one we gave in (42)b for the same sentence.

- (60)  $RS_a$   
 a. IX-a 1-WALK-WITH-ENERGY(CL-ONE).

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(i)  $[[Op S]]^{c, w} = [[S]]^{w, w}$

<sup>23</sup> Two remarks should be added.

1. It is not necessarily the case that a single viewpoint is associated with a context. If several are, the analysis will have to be refined. (This is the same problem we discussed in Section 5.4 in relation to the viewpoint shift analysis).

2. Schlenker et al. 2025 adopt the slightly more constrained definition in (i), on which the viewpoint depends only on the *agent* of  $c$ . Since in our data the viewpoint needn't be strictly that of an individual, we go with the more liberal definition in the text (the viewpoint is associated with  $c$ , but not necessarily just with the agent of  $c$ ).

(i)  $[[\pi^*]]^{c, s, t, w} = \pi(c_a)$ , the dynamic viewpoint associated with the agent of  $c$  (Schlenker et al. 2025)

'He left with energy.' (ASL, 14, 233)

a'.  $\text{pro}_a \text{RS}_a \text{person-cl}^{\pi^*}$

$$\begin{aligned} \text{b. } [[(a')]]^{c, s, t, w} &= [[\text{RS}_a \text{person-cl}^{\pi^*}]]^{c, s, t, w} ([\text{pro}_a]]^{c, s, t, w}) \\ &= [[\text{person-cl}^{\pi^*}]]^{<s(a), t, w>, s, t, w} ([\text{pro}_a]]^{c, s, t, w}) \\ &= [\lambda x_e . \text{person}'_{t, w}(x) = 1 \text{ and } \text{proj}(x, \pi(<s(a), t, w>), t, w) = \text{person-cl}](s(a)) \\ &= 1 \text{ iff } \text{person}'_{t, w}(s(a)) = 1 \text{ and } \text{proj}(s(a), \pi(<s(a), t, w>), t, w) = \text{person-cl} \end{aligned}$$

As can be seen in the derivation in (60), since the context is shifted and the context-bound viewpoint variable depends on the shifted context, iconic expressions that appear under Role Shift are evaluated with respect to the shifted viewpoint, as is desired.<sup>24</sup>

Similarly, the last sentence of (25)b can be analyzed as in (61), where the Role Shift operator has scope over the rest of the sentence.

$$(61) \text{RS}_b [\emptyset_b \text{palms-up} \& \emptyset_b \text{eyes-drop} \& \text{scared} \& \text{elephant animal-move-cl}^{\pi^*}]$$

The Role Shift operator has no effect on the first two conjuncts of (61), as they contain neither indexicals nor the context-bound viewpoint variable  $\pi^*$ . Regarding the last conjunct, which does contain  $\pi^*$ , its truth conditions are affected by context shift. To simplify matters, we symbolize the first two conjuncts as  $p$  and  $p'$  (with meanings  $\mathbf{p}$  and  $\mathbf{p}'$  respectively), and we provide in (62) the truth conditions of (61). We assume that *elephant* denotes a particular elephant, written as *elephant'*.

$$\begin{aligned} (62) [[(61)]]^{c, s, t, w} &= [[\text{RS}_b [p \& p' \& \text{elephant animal-move-cl}^{\pi^*}]]]^{c, s, t, w} \\ &= [[p \& p' \& \text{elephant animal-move-cl}^{\pi^*}]]^{<s(a), t, w>, s, t, w} \\ &= \mathbf{p}_{t, w} = 1 \text{ and } \mathbf{p}'_{t, w} = 1 \text{ and } [[\text{elephant animal-move-cl}^{\pi^*}]]^{<s(a), t, w>, s, t, w} = 1 \\ &= \mathbf{p}_{t, w} = 1 \text{ and } \mathbf{p}'_{t, w} = 1 \text{ and } [[\text{animal-move-cl}^{\pi^*}]]^{<s(a), t, w>, s, t, w} ([[\text{elephant}]]^{<s(a), t, w>, s, t, w}) \\ &= \mathbf{p}_{t, w} = 1 \text{ and } \mathbf{p}'_{t, w} = 1 \text{ and } [\lambda x_e . \text{animal}'_{t, w}(x) = 1 \text{ and } \text{proj}(x, \pi(<s(a), t, w>), t, w) = \text{animal-move-cl}](\text{elephant}') \\ &= \mathbf{p}_{t, w} = 1 \text{ and } \mathbf{p}'_{t, w} = 1 \text{ and } \text{animal}'_{t, w}(\text{elephant}') = 1 \text{ and } \text{proj}(\text{elephant}', \pi(<s(a), t, w>), t, w) = \text{animal-move-cl} \end{aligned}$$

This seems appropriate.

This analysis could be extended to non-grammatical facial expressions, such as those in (9)b and (10)b, by making them sensitive to a viewpoint of evaluation, as is natural in view of their iconic nature.

Importantly, we can provide in this framework the same kind of analysis of the blocking of indexicals under Action Role Shift as in Schlenker 2017a. The main idea was that indexicals must depend on a "rich" context, that of a speech/thought act, something that Action Role Shift does not offer, as it delivers a mere perspectival point. Schlenker 2017a implemented these ideas in terms of a type system that yields a formal difference between rich and impoverished contexts. This or alternative implementations could be adopted. The result will be the same: the analysis of Role Shift as overt context+viewpoint shift has a way to capture this generalization. (As we mentioned above, in view of data in Schlenker et al., to appear, the status of the generalization ought to be revisited in future research.)

### 6.3 Attitude Role Shift

Turning to Attitude Role Shift, we will adopt from the context-shifting analysis the idea that attitude reports quantify over contexts:  $x$  says that  $F$  is true just in case for every context  $c'$  compatible with what  $x$  says,  $c'$  makes  $F$  true. But we will add a twist to it: we will assume that these contexts  $c'$  are themselves associated with a signing event, which can be iconically represented by the role-shifted clause. (This analysis may optionally be extended to thinking events, if the thinking involves an inner monologue, or if it is interpreted metaphorically. For simplicity, we restrict attention to saying events.)

<sup>24</sup> See fn. 20 for the interpretation of  $\text{proj}(s(a), \pi(<s(a), t, w>), t, w) = \text{person-cl}$  when  $\pi(<s(a), t, w>)$  is a dynamic viewpoint.

### 6.3.1 Context shifting

We start from a standard De Se semantics for *say*, implemented by way of the syncategorematic rule in (63) (e.g. Chierchia 1989, Schlenker 2011). The embedded clause is assumed to correspond to a property of individuals.<sup>25</sup>

*Notation:* If  $c$  is a context,  $c_a$  is its agent,  $c_t$  its time and  $c_w$  its world.

- (63) If  $c$  is a context,  $s$  an assignment function,  $t$  a time and  $w$  a world, and if  $\llbracket F \rrbracket^{c, s, t, w}$  is of type  $\langle e, t \rangle$ , then  
 $\llbracket \text{say } F \rrbracket^{c, s, t, w} = \lambda x_e . \text{ for every } c' \in \text{SAY}_{t, w}(x), \llbracket F \rrbracket^{c', s, c'_t, c'_w}(c'_a) = 1$   
 where  $\text{SAY}_{t, w}(x)$  is the set of context compatible with what  $x$  says at  $t$  in  $w$ .

We can apply this lexical entry to a clause without Role Shift, for instance to the De Se reading of (64)a, from Schlenker 2017a, which involves an embedded third person pronoun. We give it the Logical Form in (64)a', which follows various accounts (including Schlenker 2017a) in positing an abstractor over individuals in the embedded clause.<sup>26</sup>

- (64) a. IX-a SAY IX-a WILL LEAVE  
 'He said he would leave.' (ASL 14, 231; Schlenker 2017a)  
 a'.  $\text{pro}_a \text{ say } \lambda a \text{ pro}_a \text{ will-leave}$   
 b.  $\llbracket (a') \rrbracket^{c, s, t, w} = \llbracket \text{think } \lambda a \text{ pro}_a \text{ will-leave} \rrbracket^{c, s, t, w}(\llbracket \text{pro}_a \rrbracket^{c, s, t, w})$   
 $= [\lambda x_e . \text{ for every } c' \in \text{SAY}_{t, w}(x), \llbracket \lambda a \text{ pro}_a \text{ will-leave} \rrbracket^{c', s, c'_t, c'_w}(c'_a) = 1](s(a))$   
 $= 1 \text{ iff for every } c' \in \text{SAY}_{t, w}(s(a)), \llbracket \lambda a \text{ pro}_a \text{ will-leave} \rrbracket^{c', s, c'_t, c'_w}(c'_a)$   
 $= 1 \text{ iff for every } c' \in \text{SAY}_{t, w}(s(a)), \text{will-leave}_{c'_t, c'_w}(c'_a) = 1$

*In words:* Every context  $c'$  compatible with what  $s(a)$  says at  $t$  in  $w$  is such that the agent of  $c'$  leaves at the time of  $c'$  in the world of  $c'$  (and hence  $c'$  makes true 'I will leave').

This analysis can be combined with our treatment of the Role Shift operator in (57) to account for a minimally different example with Role Shift and a first person pronoun in the embedded clause, as in (65)a, from Schlenker 2017a. It can be given the Logical Form in (65)a', where the Role Shift operator is found below the  $\lambda$ -abstractor in the embedded clause.

- (65) a. IX-a SAY  $\text{RS}_a$  IX-1 WILL LEAVE.  
 He said: 'I will leave.' (ASL 14, 233; Schlenker 2017a)  
 a'.  $\text{pro}_a \text{ say } \lambda a \text{ RS}_a \text{ I will-leave [1st try]}$   
 b.  $\llbracket (a') \rrbracket^{c, s, t, w} = 1 \text{ iff for every } c' \in \text{SAY}_{t, w}(s(a)), \llbracket \lambda a \text{ RS}_a \text{ I will-leave} \rrbracket^{c', s, c'_t, c'_w}(c'_a) = 1$   
 $\text{iff for every } c' \in \text{SAY}_{t, w}(s(a)), [\lambda x_e \llbracket \text{RS}_a \text{ I will-leave} \rrbracket^{c', s[a \rightarrow x], c'_t, c'_w}](c'_a) = 1$   
 $\text{iff for every } c' \in \text{SAY}_{t, w}(s(a)), \llbracket \text{RS}_a \text{ I will-leave} \rrbracket^{c', s[a \rightarrow c'_a], c'_t, c'_w} = 1$   
 $\text{iff for every } c' \in \text{SAY}_{t, w}(s(a)), \llbracket \text{I will-leave} \rrbracket^{c'_a, c'_t, c'_w, s[a \rightarrow c'_a], c'_t, c'_w} = 1$   
 $\text{iff for every } c' \in \text{SAY}_{t, w}(s(a)), \text{will-leave}_{c'_t, c'_w}(c'_a) = 1$

*In words:* Every context  $c'$  compatible with what  $s(a)$  says at  $t$  in  $w$  is such that the agent of  $c'$  leaves at the time of  $c'$  in the world of  $c'$  (and hence  $c'$  makes true 'I will leave').

The result is appropriate, and it delivers the same truth conditions as the De Se reading of the embedded clause without Role Shift, as is standard for context-shifting attitude reports (for the semantic equivalence between third person De Se pronouns and shifted first person pronouns, see for instance Schlenker 2011).

<sup>25</sup> Note that in the quantification of contexts  $c'$ , each component of  $c'$  plays a role in the evaluation of the embedded clause: the agent  $c'_a$  of  $c'$  is fed as an argument to the embedded clause, while the time  $c'_t$  and world  $c'_w$  of  $c'$  appear as parameters of evaluation of the embedded clause.

<sup>26</sup> We would obtain a De Re reading with the Logical Form in (i), where the embedded subject is not bound by the  $\lambda$ -operator  $\lambda i$ .

(i)  $\text{pro}_a \text{ say } \lambda i \text{ pro}_a \text{ will-leave}$

### 6.3.2 Iconicity

To account for the quotational effects obtained under Attitude Role Shift, we propose a simple tweak: role-shifted clauses come with the viewpoint variable  $\pi^*$ , the distinguished variable that depends on the context. This is natural because the signer's shifted position suggests that the viewpoint is shifted as well. So the Logical Form of (65)a should not be the one we posited before, repeated in (66)a, but that in (66)b, in accordance with (67).<sup>27</sup>

- (66) a. Without viewpoint shift:  $\text{pro}_a \text{ say } \lambda a \text{ RS}_a [\text{I will-leave}]$   
 b. With viewpoint shift:  $\text{pro}_a \text{ say } \lambda a \text{ RS}_a [\text{I will-leave}]^{\pi^*}$

(67) A role-shifted clause under an attitude verb must carry the distinguished variable  $\pi^*$ .

The question is how to interpret a clause that carries a viewpoint variable. We can't quite use the rule for classifiers in (5)b, copied in (68), as a role-shifted clause does not denote a property of individuals (nor is it a classifier).

- (68) If  $P$  is a 1-place predicate and if  $\pi$  is a viewpoint-denoting expression  

$$[[P^\pi]]^{c, s, t, w} = \lambda x_e . [[P]]^{c, s, t, w}(x) = 1 \text{ and } \mathbf{proj}(x, [[\pi]]^{c, s, t, w}, t, w) = P$$

But we can do something close, which dovetails with Davidson's (2015) and Supalla's (1982) idea that the body serves as a classifier in such cases. We can take a clause to denote a property of contexts—and to be true if it holds of the context of evaluation. This leads to the following definition (but see Appendix II for a slightly different derivation of the same result):

- (69) If  $S$  is a clause,  $\pi$  a viewpoint variable,  $c$  a context associated with a sign or thought event  $\text{event}(c)$ ,  $s$  an assignment function,  $t$  a time and  $w$  a world,  

$$[[S^\pi]]^{c, s, t, w} = 1 \text{ iff } [[S]]^{c, s, t, w} = 1 \text{ and } \mathbf{proj}(\text{event}(c), [[\pi]]^{c, s, t, w}, t, w) = S$$

In essence, a clause  $S$  with a viewpoint variable  $\pi$  is true in a context  $c$  just in case it is true when evaluated in that context (together with the other parameters evaluation), and in addition the sign or thought event associated with  $c$  projects to  $S$  from the viewpoint denoted by  $\pi$ .

This makes it possible to develop a more appropriate analysis of Attitude Role Shift, with a quotational component, as is illustrated in (70).

- (70) a.  $\text{pro}_a \text{ say } \lambda a \text{ RS}_a [\text{I will-leave}]^{\pi^*}$  [2nd try]  
 b.  $[[[a']]]^{c, s, t, w} = 1$  iff for every  $c' \in \text{SAY}_{t, w}(s(a))$ ,  $[[\lambda a \text{ RS}_a [\text{I will-leave}]^{\pi^*}]]^{c, s, c'_t, c'_w}(c'_a) = 1$   
 iff for every  $c' \in \text{SAY}_{t, w}(s(a))$ ,  $[\lambda x_e [[\text{RS}_a [\text{I will-leave}]^{\pi^*}]]^{c, s[a \rightarrow x], c'_t, c'_w}(c'_a) = 1$   
 iff for every  $c' \in \text{SAY}_{t, w}(s(a))$ ,  $[[\text{RS}_a [\text{I will-leave}]^{\pi^*}]]^{c, s[a \rightarrow c'_a], c'_t, c'_w} = 1$   
 iff for every  $c' \in \text{SAY}_{t, w}(s(a))$ ,  $[[[\text{I will-leave}]^{\pi^*}]]^{c'_a, c'_t, c'_w, s[a \rightarrow c'_a], c'_t, c'_w} = 1$   
 iff for every  $c' \in \text{SAY}_{t, w}(s(a))$ ,  $\text{will-leave}_{c'_t, c'_w}(c'_a) = 1$  and  $\mathbf{proj}(\text{event}(\langle c'_a, c'_t, c'_w \rangle),$   
 $[[\pi^*]]^{c'_a, c'_t, c'_w, s[a \rightarrow c'_a], c'_t, c'_w}, c'_t, c'_w) = [\text{I will-leave}]$   
 iff for every  $c' \in \text{SAY}_{t, w}(s(a))$ ,  $\text{will-leave}_{c'_t, c'_w}(c'_a) = 1$  and  $\mathbf{proj}(\text{event}(\langle c'_a, c'_t, c'_w \rangle),$   
 $\pi(c'), c'_t, c'_w) = [\text{I will-leave}]$

*In words:* Every context  $c'$  compatible with what  $s(a)$  says at  $t$  in  $w$  is such that the agent of  $c'$  leaves at the time of  $c'$  in the world of  $c'$ , and in addition the signing event associated with  $c'$  projects to the embedded clause from the viewpoint of  $c'$  at the time of  $c'$  in the world of  $c'$ .

All the contexts compatible with the agent's claim must make the embedded clause true, and in addition they must be contexts of signing events that project to the embedded clause (from the

<sup>27</sup> It is tempting to consider a Logical Form such as (i), where  $\pi^*$  appears on the entire embedded clause:

- (i)  $\text{pro}_a \text{ say } [\lambda a \text{ RS}_a \text{ I will-leave}]^{\pi^*}$

The difficulty is that  $\pi^*$  is then too high to depend on the contexts  $c'$  compatible with  $s(a)$ 's claim. But we want the iconic viewpoint relative to which the signing event corresponding to  $c'$  is represented (by the embedded clause) to be the viewpoint of the agent of  $c'$ . See, however, Appendix II for an account on which  $\pi^*$  is bound from the matrix clause and the projective condition pertains directly to the signing event produced by  $s(a)$  (rather than the signing events corresponding to the contexts compatible with  $s(a)$ 's claim).

viewpoints associated with these contexts). This means that the signer recreates in front of him a version of the signs used in these contexts. As a first approximation, this seems roughly appropriate.

Still, one should be concerned that the projective condition takes scope under the attitude verb. To obtain appropriate truth conditions, we must posit, as in (71), that if the attitude holder claims that his context has a certain form, then this is so.

- (71) If [for every  $c' \in \text{SAY}_{t,w}(s(a))$ ,  $\text{proj}(\text{event}(\langle c'_a, c'_t, c'_w \rangle), \pi(c'), c'_t, c'_w) = p$ , then  $\text{proj}(\text{event}(\langle s(a), t, w \rangle), \pi(\langle s(a), t, w \rangle), t, w) = p$

Using (71), we can infer from the boldfaced part of the last line of (70) that  $\text{proj}(\text{event}(\langle s(a), t, w \rangle), \pi(\langle s(a), t, w \rangle), t, w) = [\text{I will-leave}]$ . In Appendix II, we explore a somewhat different system that eschews the stipulation in (71), but requires binding of the embedded viewpoint variable by a viewpoint term present in the matrix clause.

Modulo these complications, we derive the quotational component that had been noted in traditional analyses (e.g. Lillo-Martin 1995) and had to be stipulated in Schlenker 2017b, but within a framework in which iconic effects and viewpoints play a more general role. In addition, since the embedded clause is syntactically integrated, it can in principle be affected by various grammatical phenomena, including wh-extraction. (As in Schlenker 2017b, further measures would be needed to allow the quotational effects to be partial. But there is no obvious reason why the projection condition should require that the embedded clause reflect *all* of the utterance corresponding to the context.)

It is worth noting that the interpretive rule in (69) needs to be combined with context shift in order to avoid a conflict between the meaning of indexicals and their quotational use. Without context shift, we could still use this rule in the overt viewpoint shift theory with the Logical Form in (72)a, but as shown in (72)b, this does not yield the right truth conditions—for lack of a context-shifting operator, the embedded first person pronoun refers to the actual speaker (boxed part), but the quotational component of the embedded clause shows instead that this pronoun refers to the attitude holder.

- (72) Failed analysis of (65)a in the overt viewpoint shift theory (without context shift)
- a.  $\text{pro}_a \text{ say } \lambda a [\text{I will-leave}]^{\pi_a}$ , where  $s(\pi_a)$  is a viewpoint associated with  $s(a)$
  - b.  $[[\langle a' \rangle]]^{c, s, t, w} = 1$  iff for every  $c' \in \text{SAY}_{t,w}(s(a))$ ,  $[[\lambda a [\text{I will-leave}]^{\pi_a}]]^{c, s, c'_t, c'_w}(c'_a) = 1$   
 iff for every  $c' \in \text{SAY}_{t,w}(s(a))$ ,  $[\lambda x_e [[\text{I will-leave}]^{\pi_a^*}]]^{c, s[a \rightarrow x], c'_t, c'_w}(c'_a) = 1$   
 iff for every  $c' \in \text{SAY}_{t,w}(s(a))$ ,  $[[[\text{I will-leave}]^{\pi_a^*}]]^{c, s[a \rightarrow x], c'_t, c'_w} = 1$   
 iff for every  $c' \in \text{SAY}_{t,w}(s(a))$ ,  $\text{will-leave}_{c_t, c_w}(\boxed{c_a}) = 1$  and  $\text{proj}(\text{event}(\langle s(a), c'_t, c'_w \rangle), [\pi_a]^{c, s, t, w}, c'_t, c'_w) = [\text{I will-leave}]$

#### 6.4 Comparison with the original context-shifting analyses of Role Shift

The present analysis can be seen as an extension of the context-shifting analysis of Role Shift developed in Schlenker 2017a,b—following insights by Quer 2005, 2013. By adding viewpoint shift to context shift, the new analysis derives what was a stipulation (and presented as such) in the original context-shifting analysis, namely the fact that role-shifted expressions are interpreted maximally iconically. From the present perspective, iconic interpretation is due to overt viewpoint shift, which forces all iconically interpretable expressions—including the entire embedded clause for Attitude Role Shift—to be interpreted relative to a shifted viewpoint.

A relatively inessential difference pertains to analytical decisions in the analysis of Role Shift. In the present theory, the Role Shift operator is just a context-shifting operator (following Anand and Nevins 2004). In Schlenker 2017a,b, it was a context abstractor, with the semantics in (73) (we have added an abstraction over times, in line with the treatment of times by analogy with worlds; in Schlenker 2017a,b, times were disregarded for reasons of simplicity). Note that the index  $i$  played no semantic role (it just specified the locus with which Role Shift was associated).

##### (73) Definition of the Role Shift operator in Schlenker 2017a,b

Let  $c$  be a context,  $s$  an assignment function and  $w$  a situation. Then for any index  $i$  and clause  $IP$ ,

$$[[\text{RS}_i IP]]^{c, s, t, w} = \lambda x'_e \lambda t'_e \lambda w'_s. [[IP]]^{<x', t', w'>, s, t', w'}$$

As a consequence, within this earlier theory, Action Role Shift as in (60)a had to be analyzed with the Logical Form in (74), where the classifier comes with a first person agreement marker treated as a first

person pronoun (written here as  $pro_{1st}$ ) interpreted relative to a shifted context (including under Action Role Shift).

(74) **Logical Form of (60)a in a system similar to Schlenker 2017a,b**

$w^* t^* pro_a \boxed{[RS_a pro_{1st} person-cl]}$

The boxed part in essence denotes a property of contexts, with the agent, time and world coordinates taken one at a time by the functional meaning obtained through (73). For this reason, the Logical Form must contain additional arguments to saturate this function. As for the index  $I$  that appears around the role-shifted clause in (74), it forced the expression to be interpreted iconically. This was explicitly a stipulation. From the present perspective, it is derived from the fact that Role Shift overtly shows that the viewpoint of evaluation has been shifted.

## 6.5 Stepping back

Our main point has been that Role Shift involves viewpoint shift, an observation that dovetails with informal descriptions but doesn't follow from earlier formal accounts. The first theory we developed (= viewpoint shift theory) was minimal in taking Role Shift to just be overt viewpoint shift. It met with two potential difficulties. First, indexicals are usually thought to be prohibited under Action Role Shift, and it is unclear how the viewpoint shift theory can account for this fact. Second, Attitude Role shift had to be treated as simple quotation, which might not account for its grammatical flexibility. The second theory we developed starts from the analysis of Role Shift as overt context shift, but with the addition that context shift involves viewpoint shift as well (= context+viewpoint shift theory). As in earlier context-shifting analyses, this makes it possible to posit that indexicals need to depend on rich contexts (corresponding to speech- or thought-acts) rather than impoverished ones (found in Action Role Shift). And as is the case in the analysis of classifiers in Iconological Semantics, it is possible to state that role-shifted clauses under Attitude Role Shift are simultaneously used and quoted. The second theory improves on earlier context-shifting analyses in offering an articulated account of viewpoint shift. It also derives the fact that Role Shift comes with a condition of maximal iconicity, something that motivated the category of Super Monsters. From the present perspective, Super Monsters are in essence context shifters with viewpoint shift.

The debate between Role Shift as viewpoint shift and Role Shift as context+viewpoint shift is not settled by these remarks. While the second theory arguably has advantages over the first, it is also more complex (especially when the issue of the scope of the projective condition is taken into account), and some of the generalizations it derives require further empirical work (pertaining to indexicals under Action Role Shift, and to the grammatical integration of Attitude Role Shift).<sup>28</sup>

## 7 The Import of Gestural Role Shift

According to our first analysis ("Role Shift as overt viewpoint shift"), the key to Role Shift is that the signer signals that he is adopting a shifted viewpoint of evaluation. According to our second analysis ("Role Shift as overt context+viewpoint shift"), the signer makes it clear that signs are evaluated under a shifted context and a shifted viewpoint. In both cases, there is a connection between markers of Role Shift (such as body rotation and eyegaze shift) and the message conveyed regarding viewpoint and possibly context shift. This is essential in view of the properties of Gestural Role Shift, which seem reminiscent of those of ASL Role Shift (as noted, a full investigation is left for future research).

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<sup>28</sup> The context+viewpoint shift theory also raises a question: what is the motivation for the condition in (59)? In the viewpoint shift theory, it made sense to posit that as long as the signer is in shifted position, the viewpoint of evaluation must be the corresponding shifted viewpoint as well—otherwise why bother to remain in shifted position? But in the context+viewpoint shift theory, things are less clear: Role Shift has two functions, namely context shift and viewpoint shift, and it is thus less clear why Logical Forms couldn't include viewpoint variables different from the context-bound variable  $\pi^*$  (e.g. in Action Role Shift). As we noted at the end of Section 6.1, this degree of freedom may be a bad thing to account for our main consultant's judgments, but it may be useful to account for our second consultant's judgments, which adhered less closely to the Viewpoint Shift Generalization in (3).

Specifically, we conjectured in Section 3.1 that under Gestural Action Role Shift, (i) unembedded gestures resembling classifier predicates should start from the signer's role-shifted position; and (ii) embedded gestures of this sort are preferably interpreted from the role-shifted perspective. In addition, we suggest in Appendix III that speaker-oriented pointing gestures are acceptable under Gestural Attitude Role Shift but not under Gestural Action Role Shift. By contrast, a non-purely iconic gesture displaying subject agreement-like behavior (such as LOOK <sup>29</sup>) can appear under Gestural Action Role Shift while originating from the speaker's role-shifted position. If confirmed, these observations would show that key restrictions on indexicals under Action Role Shift but not Attitude Role Shift (e.g. Schlenker 2017a) can be replicated in gestures.

If these conjectures are on the right track, they have theoretical import, as they suggest that whatever theoretical mechanism accounts for Role Shift in sign should have a natural counterpart in gestures. We have argued that both theories discussed in this piece—Role Shift as overt viewpoint shift, and Role Shift as overt context+viewpoint shift—fit the bill.

## 8 Conclusion

### 8.1 Main results

In sum, we have seen that extant formal analyses of Role Shift have missed a key property, which was highlighted in informal descriptions: Role Shift affects the viewpoint of evaluation of iconic expressions. This has direct consequences for the interpretation of classifier predicates Action Role Shift. In Attitude Role Shift, viewpoint shift affects the interpretation of the entire role-shifted clause, which must pictorially represent a signing event.

A formal analysis of viewpoint shift was difficult to develop before the advent of Greenberg's and Abusch's pictorial semantics, a transparent formal framework in which viewpoints play a central role. Building on pictorial semantics, Iconological Semantics offers a simple way of integrating a pictorial component to sign language semantics, with viewpoint variables in the object language. In the simplest analysis, Role Shift is just overt viewpoint shift, realized by a variable  $\pi$ , and with the condition that all iconic expressions take  $\pi$  as argument. The analysis can be refined so that the viewpoint variable  $\pi$  is replaced with a functional term  $\pi(a)$ , where  $a$  is the locus corresponding to the individual that triggers Role Shift.

The analysis of Role Shift as viewpoint shift encounters two potential difficulties, however. First, it fails to explain why indexicals are arguably blocked from appearing in Action Role Shift. Second, it fails to account for phenomena that suggest that Attitude Role Shift is not standard quotation. An alternative is to analyze Role Shift as involving context+viewpoint shift. This can be seen as a natural extension of earlier context-shifting analyses, on the assumption that the context comes with a default viewpoint (the signer's), and that shifting the context shifts this viewpoint as well. While there are arguments for the second theory, some of the empirical arguments are less than univocal, and the theory is arguably more complex than the simple viewpoint shift analysis. The debate should thus be elaborated further in future research.

Either theory is natural given that the signer determines the position of the context and associated iconic viewpoint. By shifting his body and eyegaze, the signer may thus signal that the addressee must adopt a different viewpoint, or put themselves in a different context and associated viewpoint. This leads one to expect, in line with early approaches, that there could be counterparts of Role Shift with gestures. We have sketched a way to test some non-trivial properties of Role Shift in gestures, something that should be investigated in gestural research.

Stepping back, one of the quintessential phenomena of sign language grammar, Role Shift, involves viewpoint shift. This strongly argues that iconic viewpoints are first class citizens of natural language semantics, as posited in Iconological Semantics.

<sup>29</sup> Picture credit: Bill Vicars, <https://www.lifeprint.com/asl101/pages-signs/l/look.htm>.



## 8.2 Open issues

Besides this gestural extension, several important issues are left for future sign research.

On an empirical level, while we based our discussion on the detailed and repeated judgments of our main consultant, there are non-trivial disagreements with our second consultant, who adheres less strongly to the Viewpoint Shift Generalization in (3). Our paradigms should definitely be explored with further signers, as we have solely aimed to provide the analysis of one idiolect. In addition, extension to other sign languages would be of great interest (and testing the conjectures about gestures mentioned in Section 7 would be important as well). More complex examples should be constructed to test the potential limits of the Viewpoint Shift Generalization in (3). And this generalization should be explored in further sign languages.<sup>30</sup>

On a theoretical level, there are several open questions. First, what is the proper definition of Role Shift? We used body rotation as a telltale sign that Role Shift is applied, but one might think that whenever the signer conveys—by whatever means—that another viewpoint/context is adopted, Role Shift is at work. A key issue will be to develop clear independent tests of Role Shift, for instance involving the behavior of indexicals; but this is not always an easy task (see Schlenker et al., to appear, for a recent analysis of ASL viewpoints that encountered precisely this problem; see also Janzen 2025).<sup>31</sup>

Second, it might become necessary to distinguish among different varieties of Role Shift, depending on details of realization and/or interpretation. Waller 2021 thus concludes that there is a distinction between vantage point shift (viewpoint shift, in our terms) and character Role Shift. The typology of Role Shift operations might thus have to be ramified.

Third, an important issue has been left out of the present discussion to avoid clouding the main point. Multiple authors have noted that signers can simultaneously combine multiple perspectives, for instance with a classifier and some facial expressions iconically representing different entities (e.g. Dudis 2004; Perniss 2012; Quinto-Pozos 2007, Parrill et al. 2016<sup>32</sup>). Our framework with viewpoint variables in the object language should allow for an analysis of such cases, but their formal investigation is left for future research.

Fourth, we have assumed a theory of ASL context shift in which context is a parameter, not a variable. This was because our ASL data fall under Anand and Nevins's (2004) Shift Together constraint (Schlenker 2017a, b). But as noted above (in the discussion of (55)), Shift Together arguably fails in Catalan Sign Language (LSC) and in German Sign Language (DGS). The theories of Role Shift developed in this piece would correspondingly have to be weakened to be extended to these languages. If Role Shift is analyzed as simple viewpoint shift, LSC and DGS data might suggest that some expressions can be exempted from viewpoint shift, leading in effect to mixed quotation (e.g. Maier 2017, 2018). If Role Shift is analyzed as context+viewpoint shift, the natural thought is to treat context dependency in terms of context variables rather than context parameters, as was done in original theories of context shift, before the advent of Shift Together (Schlenker 1999, 2003; see Guerrini 2022 for a revival of these ideas). In fact, the existence of both types of patterns (Shift Together vs. Mix Perspectives) across sign languages and across spoken languages has been mentioned as a similarity

<sup>30</sup> An anonymous reviewer points out an example from DGS (German Sign Language) in which Role Shift to an individual (a shepherd boy) co-occurs with facial expressions attributed to another character (the shepherd boy's neighbors). This example (Maier and Steinbach 2022, example (15)) violates the Viewpoint Shift Generalization in (3) (stated only for ASL!), and thus both theories developed in the present piece fail to capture it. But the mechanisms of Iconological Semantics—though not our specific theories—could in principle capture the reviewer's example. The reason is that the formalism of Iconological Semantics allows (i) several viewpoint variables to co-occur in the same clause, and (ii) context shift to take place without affecting all viewpoints of the affected expressions. But the facts need to be systematically investigated before an analysis can be developed.

<sup>31</sup> The strict criteria adopted in this piece imply that many examples without body rotation but with emotional facial expressions attributed to a character (e.g. *excitedly* in (18)a-c) do *not* qualify as Role Shift. This could be revisited in the future, but ours seems to be the right definitional move, as such examples differ from those with body rotation in not forcing viewpoint shift. In other words, if there is Role Shift, emotional facial expressions are attributed to the agent (as in (10)b), but not conversely.

<sup>32</sup> For instance, Parrill et al. 2016 write that in sign language, "non-manual signals may combine with manuals to present multiple perspectives of an entity or event (Dudis 2004; Perniss 2012; Quinto-Pozos 2007)".

between grammar in the visual modality and grammar in the vocal modality (Schlenker 2017a,b, 2018a). Hübl et al. 2019 discuss in detail the theoretical consequences of the DGS pattern, and mention two further possibilities besides the appeal to context variables—the use of operators that shift different coordinates of the context parameter (Deal 2020), and mechanisms of partial quotation (Maier 2017, 2018).

Finally, our results on ASL should be integrated to a more general typology of viewpoint- and context-sensitive operations in language. Schlenker and Lamberton 2024 argued that the viewpoint-dependency they found in ASL classifier constructions has counterparts in English gestures, and that the latter tend to correlate with the perspective-dependency of expressions like *come* and *go*. This might suggest that a unified notion of viewpoint-dependency is at work in all these constructions. If so, one might ask about operators that shift the viewpoint (e.g. of *come* and *go*) in spoken language, and also about correlations between context shift and viewpoint shift in spoken languages that have Kaplanian monsters.<sup>33</sup>

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<sup>33</sup> One might specifically ask (as did Schlenker 2017a p. 56) whether the behavior of iconic elements under Role Shift has parallels in context-shifting constructions in spoken language.

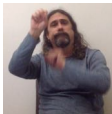
## *Appendix I. Additional Examples*

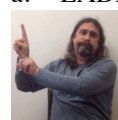
We provide below additional data on Action and Attitude Role Shift.

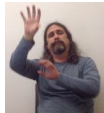
### A. Additional Data on Action Role Shift

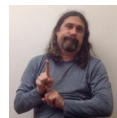
The following example of Action Role Shift presents an exception to the generalization we saw in the main text. A ladder is represented on the signer's right. Mary climbing down the ladder can be represented on the right as well (keeping the original viewpoint from which the ladder was depicted in the context), or in front of the signer, if viewpoint shift is applied. Since Role Shift normally implies viewpoint shift, we would expect that (75)c should be degraded because it involves Role Shift to Mary's perspective, but no corresponding viewpoint shift, as the classifier representing Mary starts out high in signing space. By contrast, (75)d should be more acceptable because the classifier representing Mary starts roughly from the signer's position. For our main consultant, the data do not bear this out: (75)c is almost as acceptable as (75)d. (Our second consultant finds a slight difference, scoring (75)c as 6 and (75)a,b,d as 7.)

- (75)  YESTERDAY RENOVATION. LADDER\_ JOHN<sub>high-right</sub> HIGH, MARY<sub>high-right</sub> TOP. JOHN<sub>high-right</sub> LAZY, SO MARY<sub>high</sub>  
'Yesterday, there was some renovation work. John and Mary were on a ladder, with John high and Mary at the top. John was lazy, so Mary

- a. <sup>6,7</sup> LADDER-CLIMB-DOWN\_  high-right-PERSON-cl-medium-right^PERSON-cl<sub>medium-right</sub>\_



- b. <sup>6,7</sup> LADDER-CLIMB-DOWN\_  medium-front-PERSON-cl-low-front^PERSON-cl<sub>low-front</sub>\_

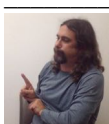


- c. <sup>6,7</sup> RS<sub>right</sub>\_\_\_\_\_

- LADDER-CLIMB-DOWN\_  high-right-PERSON-cl-medium-right^PERSON-cl<sub>medium-right</sub>\_



- d. <sup>7</sup> RS<sub>right</sub>\_\_\_\_\_
- LADDER-CLIMB-DOWN\_  medium-front-PERSON-cl-low-front^PERSON-cl<sub>low-front</sub>\_



started climbing down the ladder, hitting John.'  
(ASL 37, 3424, 3 judgments; <https://youtu.be/82bsFwjpM2Q>)

We believe that (75)c is acceptable for our main consultant because Role Shift signals two successive viewpoint shifts, first to adopt a viewpoint associated with Mary climbing down, then to adopt the viewpoint of a person observing the scene from the side of the ladder. Both viewpoints are licensed by the signer's rotation towards the right.<sup>34</sup> So in this case as well, Role Shift implies viewpoint shift, but in a more complicated fashion than in the examples discussed in the main text.

## **B. Additional Data on Attitude Role Shift**

For completeness, we discuss counterparts with Attitude Role Shift of examples with Action Role Shift analyzed in the main text, involving the vertical dimension or complex constructions.

### **1. Attitude Role Shift and viewpoint shift: vertical dimension**

The paradigm in (76) is constructed by analogy with (22), but with an attitude report replacing an action report. Role Shift co-occurs with viewpoint shift, as seen by the deviance of (76)c. The shifted viewpoint is associated with the agent of the reported action, but not with the context of the reported speech act, as Ann is certainly not making her claim while about to jump out of a plane.<sup>35</sup> On the other hand, something close to Ann's very signs are iconically reproduced since Attitude Role Shift has a strong quotational component. Direct discourse displays a preference for the viewpoint of the agent, but the preference is weak. We do not know why (77)a is not as degraded as (76)c—this does not follow from our analysis. (The puzzle is even more striking for our second consultant, who finds (76)c a bit degraded but prefers (77)a to (77)b.)

- (76) YESTERDAY ANN GO PARACHUTE. PLANE PLANE-cl<sub>circling a</sub>. PLANE DOOR-OPEN. ANN<sub>a</sub>  
'Yesterday Ann went parachuting. The plane circled, the door opened, and Ann

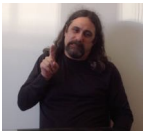
RECENTLY SAY

recently said

a. <sup>6.7</sup> hesitant\_\_\_\_\_

IX-a **high-a-PERSON<sub>high-cl</sub>**  **high-a-PERSON-falling-low-cl.**  
that she hesitated and jumped.'

b. <sup>6.3</sup> hesitant\_\_\_\_\_

IX-a **1-PERSON-cl**  **1-PERSON-falling-low-cl.**

<sup>34</sup> This analysis is suggested by the consultant's detailed answers to the following questions (see the Supplementary Materials-B, [JL 24.10.16]); the underlined parts are indicative of a shift away from the shifted viewpoint associated with Mary.

(i) What is/are the position(s) from which the addressee observes the scene? "from side of the ladder, near then far"

(ii) What is/are the position(s) that the signer adopts to represent the scene? "Mary, then ground near front of ladder"

<sup>35</sup> See the remark in fn. 16.

that she hesitated and jumped.'

c. <sup>3.3</sup> RS<sub>a</sub> \_\_\_\_\_  
 hesitant \_\_\_\_\_



IX-1 high-a-PERSON<sub>high-cl</sub> **high-a-PERSON-falling-low-cl.**  
 : I hesitated and jumped.'

d. <sup>7</sup> RS<sub>a</sub> \_\_\_\_\_  
 hesitant \_\_\_\_\_



IX-1 1-PERSON-cl\_ **1-PERSON-falling-low-cl.**

: I hesitated and jumped.'

(ASL 37, 3440, 3 judgments; <https://youtu.be/dS8yLrBUTbA>)

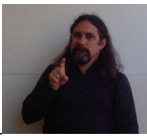
(77) *Context:* Yesterday the signer went sky diving. The plane circled and circled. The door opened...

hesitant \_\_\_\_\_



a. <sup>6</sup> IX-1 PERSON<sub>high-cl</sub> **high-a-PERSON-falling-low-cl.**

hesitant \_\_\_\_\_



b. <sup>7</sup> IX-1 1-PERSON-cl\_ **1-PERSON-falling-low-cl.**

'I hesitated and jumped.'

(ASL 37, 3476, 3 judgments; <https://youtu.be/8039bxDb0hM>)

## 2. Attitude Role Shift and viewpoint shift: complex constructions

We turn to an Attitude Role Shift counterpart of an additional complex construction discussed in Section 3.4. It dovetails with our discussion of Section 4.1. Our paradigm appears in (78), with the classifier movement in (79) (we provide a depiction of Ann to help interpret the diagram). Without Role Shift, an ambiguity is obtained, as in (78)a,b (without the surprised facial expression, as in (78)a, a reading without viewpoint shift is preferred). With Role Shift, as in (78)c,d, the embedded clause is understood with viewpoint shift. Importantly, the shifted viewpoint is that of the agent of the reported action, not of the attitude holder. (Our second consultant found an ambiguity relative to viewpoint shift in (78)a-b, and seemed to apply viewpoint shift in (78)c-d; see the Supplementary Materials C.)

(78) *Context:* Ann often tells surprising stories about the signer's cat or hers. These stories may or may not be true, depending on the case.

YESTERDAY POSS-1 CAT IN CLASS. ANN<sub>b</sub> TEACHER EYES ITCH. STUDENT b-ROW-a-cl.  
 ANN<sub>b</sub> EXPLAIN-a PERSON<sub>oscilating, left hand-cl</sub><sub>b</sub>. LATER IX-b SAY  
 'Yesterday, my cat was in class. Ann was the teacher, and her eyes itched. The students were in rows in front of her, and she explained things to them. Later, she said that as she  
 a. <sup>6</sup> TEACH PALMS-UP EYES-DROP APPEAR POSS-1 CAT a-ANIMAL-b-cl.  
 taught, she put in eye drops, and there appeared my cat, which moved (i) [unlikely] diagonally towards Ann, back to front, right to left, or (ii) [likely] towards Ann.

b. <sup>6.7</sup> TEACH PALMS-UP EYES-DROP :o) APPEAR POSS-1 CAT a-ANIMAL-b-cl.

taught, she put in eye drops. And then she was surprised to see that my cat moved **(i) diagonally towards Ann, back to front, right to left, or (ii) towards Ann.**

c. RS<sub>b</sub>

<sup>6.7</sup> TEACH PALMS-UP EYES-DROP APPEAR POSS-1 CAT a-ANIMAL-b-cl.

taught, she put in eye drops, and there appeared her cat, which moved **in front of her, right to left.**

d. RS<sub>b</sub>

<sup>6.7</sup> TEACH PALMS-UP EYES-DROP :o) APPEAR POSS-1 CAT a-ANIMAL-b-cl.

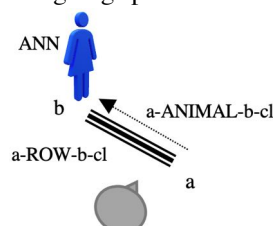
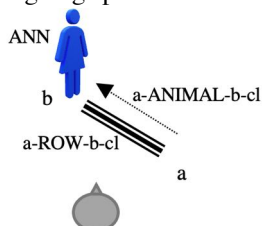
taught, she put in eye drops. And then she was surprised to see that her cat moved **in front of her, right to left.**

POSS-b STORY TRUE NOT-KNOW.

I don't know whether what she said is true.'

(ASL 37, 3296, 3 judgments; [https://youtu.be/eMiNrx\\_LTU](https://youtu.be/eMiNrx_LTU))

(79) a. Signing space without Role Shift in (27)a,b b. Signing space with Role Shift in (27)c,d



## Appendix II. A Modification of the Context+Viewpoint Shift Theory

In the main text, our account of Attitude Role Shift had the projective (quotational) component take scope under the attitude report, as was illustrated in (70). For a sentence of the form *Ann says RS p*, this yields in essence: Ann says (i) that *p*, and (ii) that she uses the very words in *p*. One might think that (ii) is undesirable and that there should be, directly, an inference that Ann uses the very words in *p*. We sketch an alternative in which the projective condition is evaluated outside the scope of the attitude operator thanks to a viewpoint term that appears in the matrix clause.

For simplicity, we consider only a framework with static viewpoints (provisions would be needed to take into account dynamic viewpoints, as in Schlenker et al., to appear). There are two key ideas. First, static viewpoints are redefined so that as to be Lewisian objects, i.e. objects situated at a time and in a world<sup>36</sup> (in Schlenker and Lamberton 2024, where issues of intensionality didn't matter, viewpoints were in essence Cartesian frames of reference, which provided spatial but not temporal or modal information). Second, the distinguished viewpoint variable  $\pi^*$  that appears under Role Shift must be bound by a functional term  $\pi(a)$  that returns the (static, Lewisian) viewpoint associated with  $s(a)$  (the individual denoted by  $a$ ) at the time and in the world of evaluation. On the assumption that the term  $\pi(a)$  lies outside the scope of the attitude operator, it will contain enough information to provide a projective condition for Attitude Role Shift that pertains to the actual saying event that took place at the matrix time and world of evaluation.

Key modified assumptions about viewpoints appear in (80). (Projection can be dynamic, as in Schlenker and Lamberton 2024, to account for the information provided by dynamic classifier predicates.)

### (80) Modified assumptions about viewpoint variables and viewpoint terms

- (i) Static viewpoints are Lewisian objects, i.e. they provide not just spatial information but also determine a time and a world.
- (ii) If  $i$  is any index,
  - a.  $[[\pi_i]]^{c, s, t, w} = s(\pi_i)$  (a static viewpoint, which is Lewisian, i.e. specifies a time and world)
  - b.  $[[\pi^*]]^{c, s, t, w} = s(\pi^*)$  (same remark)
  - c.  $\pi^*$  must be bound by a functional viewpoint term of the form  $\pi(i)$
  - d.  $[[\pi(i)]]^{c, s, t, w} = \pi(s(i), t, w)$ , the static (Lewisian) viewpoint associated with  $s(i)$  at  $t$  in  $w$ .

In view of (80)c, Action Role Shift can be analyzed as in (81)a",<sup>37</sup> which gives rise to the same kind of truth conditions as before. (Further constraints would be needed to prevent the system from overgenerating; we do not attempt to state such constraints here.)

### (81) Revised analysis of Action Role Shift

- RS<sub>a</sub>\_\_\_\_\_
- a. IX-a 1-WALK-WITH-ENERGY(CL-ONE).  
'He left with energy.' (ASL, 14, 233)
- a'. Old LF:  $\text{pro}_a \text{RS}_a \text{person-cl}^{\pi^*}$
- a". Revised LF:  $\pi(a) \lambda \pi^* \text{pro}_a \text{RS}_a \text{person-cl}^{\pi^*}$
- b.  $[[[a'']]^{c, s, t, w} = [[\text{pro}_a \text{RS}_a \text{person-cl}^{\pi^*}]]^{c, s[\pi^* \rightarrow \pi(s(a), t, w)], t, w}$   
 $= [[[\text{RS}_a \text{person-cl}^{\pi^*}]]^{c, s[\pi^* \rightarrow \pi(s(a), t, w)], t, w} ([[\text{pro}_a]]^{c, s[\pi^* \rightarrow \pi(s(a), t, w)], t, w}$   
 $= [[[\text{person-cl}^{\pi^*}]]^{c, s(a), t, w}, s[\pi^* \rightarrow \pi(s(a), t, w)], t, w} ([[\text{pro}_a]]^{c, s[\pi^* \rightarrow \pi(s(a), t, w)], t, w}$   
 $= [\lambda x_e . \text{person}'_{t, w}(x) = 1 \text{ and } \text{proj}(x, \pi(s(a), t, w), t, w) = \text{person-cl}](s(a))$   
 $= \text{person}'_{t, w}(s(a)) = 1 \text{ and } \text{proj}(s(a), \pi(s(a)), t, w), t, w) = \text{person-cl}$

To extend this analysis to Attitude Role Shift, we must redefine the interpretation of clauses carrying the distinguished variable  $\pi^*$ . The latter will now provide information about the saying event

<sup>36</sup> See for instance Lewis 1986. Semantic systems are often based on trans-world and trans-time individuals, whereby the same individual exists in several possible worlds and times. By contrast, Lewisian individuals are bound to a time and world, and have counterparts at other times and in other worlds.

<sup>37</sup> (81)a" could be taken to arise from the Logical Form in (i)a, followed by movement of  $\pi(a)$ , yielding (i)b.

- (i) a.  $\text{pro}_a \text{RS}_a \text{person-cl}^{\pi(a)}$   
b.  $\pi(a) \lambda \pi^* \text{pro}_a \text{RS}_a \text{person-cl}^{\pi^*}$

that must iconically project to the embedded clause. The fact that viewpoints are Lewisian objects (situated at a time and in a world) is essential so we can retrieve a time and world from a viewpoint.

- (82) If  $S$  is a clause and if  $s(\pi^*)$  is a (Lewisian, static) viewpoint associated with a saying event,<sup>38</sup>  
 $\llbracket S^{\pi^*} \rrbracket^{c, s, t, w} = 1$  iff  $\llbracket S \rrbracket^{c, s, t, w} = 1$  and  $\text{proj}(\text{saying-event}(s(\pi^*)), s(\pi^*), \text{time}(s(\pi^*)), \text{world}(s(\pi^*))) = S$   
 where  
 $\text{saying-event}(s(\pi^*))$  returns the saying event associated with  $s(\pi^*)$   
 $\text{time}(s(\pi^*))$  returns the time associated with  $s(\pi^*)$   
 $\text{world}(s(\pi^*))$  returns the world associated with  $s(\pi^*)$

On the assumption that the functional viewpoint term  $\pi(a)$  appears in the matrix clause, it will provide the embedded clause with information about the agent's saying event, as shown in (83)a".<sup>39</sup> Combined with (82), this will yield a requirement that this saying event project to (i.e. be iconically represented by) the embedded clause, as shown in (83)b.

- (83)  $RS_a$  \_\_\_\_\_  
 a. IX-a SAY IX-1 WILL LEAVE.  
 He said: 'I will leave.' (ASL 14, 233; Schlenker 2017a)  
 a'. Old LF:  $\text{pro}_a \text{ say } \lambda a \text{ } RS_a \text{ [I will-leave]}^{\pi^*}$   
 a". Revised LF:  $\pi(a) \lambda \pi^* \text{ pro}_a \text{ say } \lambda a \text{ } RS_a \text{ [I will-leave]}^{\pi^*}$   
 b.  $\llbracket (a'') \rrbracket^{c, s, t, w} = 1$  iff  $\llbracket \text{pro}_a \text{ say } \lambda a \text{ } RS_a \text{ [I will-leave]}^{\pi^*} \rrbracket^{c, s[\pi^* \rightarrow \pi(s(a), t, w)], t, w} = 1$   
 iff for every  $c' \in \text{SAY}_{t, w}(s(a))$ ,  $\llbracket \lambda a \text{ } RS_a \text{ [I will-leave]}^{\pi^*} \rrbracket^{c', s[\pi^* \rightarrow \pi(s(a), t, w)], c'_t, c'_w}(c'_a) = 1$   
 iff for every  $c' \in \text{SAY}_{t, w}(s(a))$ ,  $\llbracket \lambda x_e \llbracket RS_a \text{ [I will-leave]}^{\pi^*} \rrbracket^{c', s[\pi^* \rightarrow \pi(s(a), t, w), a \rightarrow x], c'_t} \llbracket (a') \rrbracket^{c'_w}(c'_a) = 1$   
 iff for every  $c' \in \text{SAY}_{t, w}(s(a))$ ,  $\llbracket RS_a \text{ [I will-leave]}^{\pi^*} \rrbracket^{c', s[\pi^* \rightarrow \pi(s(a), t, w), a \rightarrow x], c'_t, c'_w} = 1$   
 iff for every  $c' \in \text{SAY}_{t, w}(s(a))$ ,  $\llbracket \text{[I will-leave]}^{\pi^*} \rrbracket^{c'_a, c'_t, c'_w, s[\pi^* \rightarrow \pi(s(a), t, w), a \rightarrow x], c'_t, c'_w} = 1$   
 iff for every  $c' \in \text{SAY}_{t, w}(s(a))$ ,  $\text{will-leave}_{c'_t, c'_w}(c'_a) = 1$  and  **$\text{proj}(\text{saying-event}(\pi(s(a), t, w)), \pi(s(a), t, w), \text{time}(\pi(s(a), t, w)), \text{world}(\pi(s(a), t, w))) = \text{[I will-leave]}$**

*In words:* Every context  $c'$  compatible with what  $s(a)$  says at  $t$  in  $w$  is such that the agent of  $c'$  leaves at the time of  $c'$  in the world of  $c'$ , and the saying event associated with  $s(a)$  at  $t$  in  $w$  projects to the embedded clause.

The boldfaced condition on the last line of (83)b pertains to the  $s(a)$ 's actual saying event, as is desired.<sup>40</sup> Obviously, the details of this analytical direction would need to be spelled out and justified.

<sup>38</sup> In a full account, there should be a presupposition that  $s(\pi^*)$  is associated with a saying event, but we leave this out of this simplified presentation.

<sup>39</sup> Here too, (83)a" could be taken to arise from the Logical Form in (i)a, followed by movement of  $\pi(a)$ , yielding (i)b. But note that the embedded  $\lambda$ -operator must introduce a variable  $a'$  rather than  $a$  to avoid unduly binding the argument of  $\pi(a)$ .

- (i) a.  $\text{pro}_a \text{ say } \lambda a' \text{ } RS_{a'} \text{ [I will-leave]}^{\pi(a)}$   
 b.  $\pi(a) \lambda \pi^* \text{ pro}_a \text{ say } \lambda a' \text{ } RS_{a'} \text{ [I will-leave]}^{\pi^*}$

<sup>40</sup> Let us add that this boldfaced condition is only sensitive to parameters of the matrix clause (through the dependency on  $\pi(a)$ ). Since the right-hand side of the projective condition is only sensitive to overt material, one might consider positing Logical Forms such as (i)a or (i)b, where  $\pi^*$  appears on a larger constituent of the embedded clause. Interpretive rules for clausal material carrying a viewpoint variable would have to be adapted, however.

- (i) a.  $\pi(a) \lambda \pi^* \text{ pro}_a \text{ say } \lambda a \text{ } [RS_a \text{ I will-leave}]^{\pi^*}$   
 b.  $\pi(a) \lambda \pi^* \text{ pro}_a \text{ say } [\lambda a \text{ } RS_a \text{ I will-leave}]^{\pi^*}$



### Appendix III. Gestural Role Shift and Indexicals

We strengthen the analogy between Gestural Role Shift and ASL Role Shift by considering the role of agreement-like markers vs. pointing gestures in Gestural Action Role Shift and Gestural Attitude Role Shift. We base our conjecture on our introspective judgments and those of a native speaker of American English, but we leave a verdict for future consultant-based or experimental research.

In ASL, it has been argued that a full first person pronoun can appear under Attitude Role Shift but not under Action Role Shift (Schlenker 2017a). An alternative generalization (Davidson 2015, Schlenker 2017a) is that non-iconic elements cannot appear under Action Role Shift (this is compatible with the LSF data in Schlenker 2017a, though not with the corresponding ASL data). Can similar facts—irrespective of the precise generalization—be found in gestures?

To start answering, we consider in (84) a richer variant of (11)(ii). In the absence of Gestural Role Shift, pointing towards the Robin locus on the right can co-occur with a slapping gesture targeting the Francis locus, as in (84)b. But when Gestural Role Shift is added, as in (84)e, this is less acceptable. One possibility is that the pointing gesture towards the (role-shifted) speaker doesn't add anything to the meaning, and is thus ruled out by considerations of economy. But when the pointing gesture appears outside Role Shift, as in (84)d, the result is not as unacceptable.

(84) I was standing next to IX-hand-a

[little Robin] and IX-hand-b [little Francis], and IX-hand-b [Francis]

was holding a really yummy chocolate bar but didn't want to share it.

[Alternative to (ii) in (11)]

(ii') Next thing I know,

a. SLAP-b.

b. IX-a SLAP-b.

c. RS<sub>a</sub> SLAP-b.

d. IX-a RS<sub>a</sub> SLAP-b.

e. <#>RS<sub>a</sub> IX-1 SLAP-b.

<https://youtu.be/pbDfhe11wpY>

It is worth noting that under Gestural Attitude Role Shift, the first person pointing gesture can be used, as in the following exchange (and also in (11)(i) in the main text). The meaning seems to us to be clearly quotational.

(85) I was standing next to IX-hand-a [little Robin] and IX-hand-b [little Francis], and IX-hand-b [Francis]

was holding a really yummy chocolate bar but didn't want to share it. I tried to talk to them, but both were eating marshmallows so they had to answer in gestures. I asked: What are you each going to do? And so IX-hand-b [Francis] goes:

RS<sub>b</sub> IX-1 EAT.

And IX-hand-a [Robin] goes:

RS<sub>a</sub> IX-1 PUNCH-b.

<https://youtu.be/DyYtMxNOmZQ>

It remains to ask whether first person markers are acceptable under Action Role Shift if they appear on verbs. This was the generalization found in ASL, where it is only full first person pronouns that are prohibited under Action Role Shift: first person agreement verbs are acceptable. While gestures are of course not signs, we can find gestural verbs that have two key properties of ASL agreement verbs: (i) their form is not purely iconic; (ii) their form originates in or targets different gestural loci depending on their arguments (see also Schlenker and Chemla 2018 for further similarities between agreement in some gestural verbs and in sign language agreement verbs).

A gestural verb with the meaning of *looking* fits the bill: it is not purely iconic, and its origin is the agent of the action. Its first person form (originating from the signer's position) can clearly appear under Action Role Shift, as illustrated in (86).

(86) I was standing next to IX-hand-a [little Robin] and IX-hand-b [little Francis], and IX-hand-b [Francis]

was holding a really yummy chocolate bar. Robin was on a strict diet, so IX-hand-a [Robin]

RS<sub>a</sub> 1-LOOK-b.

And IX-hand-b [Francis]:

RS<sub>b</sub> EAT.

<https://youtu.be/RDVmpPrBnMk>

These examples only scratch the surface of a rich empirical field. As in other cases in which historically unrelated sign languages converge on the same expressive means, a natural question is whether sign language Role Shift has gestural counterparts. The old idea that Role Shift resembles strategies of enactment in gestures is an essential starting point, but one must ask whether detailed properties of Role Shift can be found in gestures. Viewpoint shift is a good candidate, and the behavior of indexicals (reduced here to the speaker-directed pointing gesture and to the emblem for *look*) might be another.

If these facts were confirmed, they would bear on the cognitive origins of Role Shift in sign language, as they would suggest that non-signers can discover "on the fly" several of its properties. If so, one possibility is that these are fixed by "Universal Grammar" (UG) and thus known without prior exposure. But an alternative, in line with early approaches to Role Shift, is that its properties have deeper cognitive sources. Both theories developed in this piece (Role Shift as viewpoint shift, or as context+viewpoint shift) could in principle be made compatible with this cognitive analysis, since the notion of viewpoint is certainly central to any pictorial representation (Greenberg 2013, 2021, Abusch 2020, Abusch and Rooth 2023), and the notion of context to any communicative act. The debate between the UG-based theory and the cognitive theory should be pursued in future research.

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